

From Neuroscience to Neurotechnology: Evidence-Based Translation of Targeted Memory Reactivation into a First-Generation Closed-Loop Reference Pathway

Vahe Gdlyan^{1*}

¹Independent Researcher, Yerevan, Armenia.

Corresponding author(s). E-mail(s): gdlyanvahe31@gmail.com;

Abstract

Targeted Memory Reactivation (TMR), in which sensory cues associated with previous learning are re-presented during sleep, can influence subsequent memory performance under controlled experimental conditions. Meta-analytic evidence indicates a modest overall benefit, with an average effect size of Hedges' $g = 0.29$. Translating this experimental paradigm into practical neurotechnology nevertheless remains difficult because laboratory efficacy does not by itself establish the sensing, timing, uncertainty-management, and feedback capabilities required for automated closed-loop intervention.

This study presents a structured evidence-to-engineering analysis of TMR, integrating findings from sleep-dependent memory consolidation, TMR experiments, wearable sleep monitoring, real-time state estimation, and closed-loop stimulation. The analysis compares EEG-based and peripheral-sensor-based pathways according to physiological relevance, temporal suitability, evidence maturity, and the number of unresolved assumptions introduced into a first-generation implementation.

Peripheral sensing has demonstrated proof-of-concept feasibility for automated home TMR, but its intervention-specific control reliability has not been directly established as equivalent to electrophysiological monitoring, and the available behavioral evidence remains limited. Wearable EEG introduces greater measurement and usability burdens, yet preserves closer continuity with the physiological signals and experimental conditions used in established TMR research. Home-based EEG-guided closed-loop TMR has also demonstrated proof-of-concept feasibility and measurable behavioral benefit, although end-to-end reliability and generalizability remain unverified.

On this basis, an EEG-guided, stage-aware, real-time closed-loop TMR software layer intended for integration with compatible existing EEG hardware is identified as the most scientifically defensible first-generation pathway. This conclusion represents an evidence-bounded development hypothesis rather than a validated product or proof of technological superiority. The principal contribution of the study is a traceable methodology for translating neuroscientific evidence into system-level requirements while preserving the distinction between established findings, translational inference, and assumptions requiring subsequent engineering and experimental validation.

Keywords: Targeted Memory Reactivation, Memory Consolidation, Sleep Neuroscience, Closed-Loop Systems, Electroencephalography, Wearable Neurotechnology, Translational Neuroscience, Evidence-to-Engineering Translation

1 Introduction

Newly acquired memories are not fixed at the moment of encoding. They remain susceptible to interference and undergo post-learning processes that stabilize, reorganize, and integrate them into longer-term representations. Sleep supports this transformation through coordinated neural activity during non-rapid eye movement (NREM) sleep, including the temporal interaction of slow oscillations, sleep spindles, and hippocampal sharp-wave ripples (Diekelmann and Born 2010; Rasch and Born 2013; Klinzing et al. 2019). These processes contribute to the reactivation and reorganization of recently encoded information rather than merely protecting memory from continued waking interference.

Targeted Memory Reactivation (TMR) provides an experimental means of influencing this endogenous consolidation process. In a typical TMR paradigm, sensory cues are associated with specific learning material during wakefulness and are subsequently re-presented during sleep to bias reactivation toward the corresponding memories (Oudiette and Paller 2013). Across experimental paradigms, appropriately delivered cues have produced a modest average benefit for later memory, with a meta-analytic effect size of Hedges' $g = 0.29$. The magnitude and reliability of this effect nevertheless vary with memory domain, encoding quality, cueing protocol, and physiological context (Hu et al. 2020). TMR should therefore be understood as a condition-dependent modulation of memory consolidation rather than as a general or guaranteed method of memory enhancement.

The principal translational difficulty is that experimental efficacy does not by itself define a practical intervention system. Laboratory TMR commonly depends on physiological recording, controlled cue timing, continuous observation, and the ability to suspend stimulation when sleep becomes unstable. A system intended to operate outside this environment must reproduce the intervention-relevant functions of that workflow rather than merely generate a retrospective description of the night. It must acquire physiological information during ongoing sleep, determine whether the current interval is suitable for cueing, execute stimulation with bounded and measurable timing, and evaluate whether the cue has disrupted the state it was intended to exploit.

Recent studies provide proof-of-concept evidence that portions of this process can be automated in home environments. SleepStim used smartwatch-derived movement and cardiac information to guide auditory cue delivery, demonstrating the feasibility of peripheral-sensor-based automated TMR while also showing that outcomes depend on stimulation conditions and the avoidance of sleep disruption (Whitmore et al. 2022). A subsequent study used a wearable EEG headband to deliver vocabulary cues during detected slow-wave activity and reported a benefit for cued material under home conditions (Salfi et al. 2025). These studies narrow the distance between laboratory TMR and wearable implementation, but they do not establish that all sensing approaches provide equivalent control reliability or behavioral efficacy.

A critical distinction therefore remains between general sleep assessment and intervention-ready physiological monitoring. A sensing modality may classify sleep adequately in retrospective analysis while lacking the continuous signal access, causal state estimation, uncertainty handling, cue synchronization, or post-stimulation monitoring required for real-time closed-loop operation. Wearable EEG and peripheral sensing, including photoplethysmography (PPG), should consequently be evaluated not only by their agreement with conventional sleep staging, but by whether they can support the physiological decisions on which effective and interpretable TMR depends. The relevant translational problem is not whether the necessary components exist independently, but whether the available evidence supports their combination within a coherent first-generation pathway without converting unresolved assumptions into system requirements.

This paper addresses the following central research question:

How can current evidence on sleep-dependent memory consolidation, Targeted Memory Reactivation, and wearable sleep monitoring be translated into a scientifically defensible first-generation pathway for real-time closed-loop TMR?

To answer this question, the study develops and applies a structured evidence-to-engineering methodology. Biological and experimental findings are first translated into system-level requirements, after which candidate EEG-based and peripheral-sensor-based implementation pathways are evaluated according to physiological relevance, temporal suitability, evidence maturity, and validation risk. The intended contribution is not a new biological mechanism, engineering algorithm, or validated product. It is a traceable translational analysis that distinguishes established findings from unresolved implementation assumptions and identifies the first-generation direction most strongly justified for subsequent investigation.

The remainder of the paper is organized as follows. Section 2 defines the research scope and evidence-to-engineering methodology. Section 3 introduces the neuroscientific mechanisms required to interpret sleep-dependent memory processing, and Section 4 examines the biological rationale and experimental evidence for TMR. Section 5 evaluates wearable sleep monitoring in relation to real-time intervention requirements. Section 6 translates the combined evidence into system-level requirements and compares first-generation implementation pathways. Sections 7 and 8 define the limitations and future research directions, respectively, and Section 9 presents the final conclusions.

2 Research Scope and Methodology

2.1 Research Objective and Translational Framework

The objective of this study was to determine whether the available evidence supports a scientifically defensible first-generation pathway for real-time closed-loop Targeted Memory Reactivation (TMR). Its principal methodological contribution is a traceable process for translating findings from neuroscience, sleep research, wearable sensing, and closed-loop intervention into system-level requirements and a justified implementation direction.

The analysis distinguished four translational levels that are related but not interchangeable:

1. **Biological plausibility:** whether the proposed intervention is consistent with established mechanisms of memory formation, sleep-dependent consolidation, sensory processing, and neural reactivation;
2. **Experimental efficacy:** whether the intervention produces measurable effects under controlled conditions;
3. **Technical feasibility:** whether available sensing, computation, and intervention technologies can reproduce the functions required by the experimental paradigm; and
4. **Product readiness:** whether an integrated system demonstrates sufficient reliability, safety, usability, and generalizability for deployment.

These levels define the translational boundary of the study. Biological plausibility does not establish experimental efficacy; experimental efficacy does not guarantee technical feasibility; and technical feasibility does not demonstrate product readiness. Conclusions were therefore restricted to the level directly supported by the reviewed evidence.

The investigation integrated six evidence domains:

1. mechanisms of learning, memory formation, and consolidation;
2. sleep-dependent memory processing and neural reactivation;
3. experimental evidence for TMR;
4. wearable and portable sleep-monitoring technologies;
5. real-time sleep-state estimation; and
6. closed-loop physiological intervention.

These domains form a constraint chain rather than an independent checklist. Memory and sleep neuroscience define the physiological conditions relevant to TMR. Measurement research determines whether those conditions are observable through a candidate sensing modality. Real-time analysis determines whether the required information can be obtained during ongoing sleep, while closed-loop research determines whether that information can govern cue delivery without unacceptable disturbance. The evidence supporting each stage consequently limits the claims that can be made about the stages that follow it.

Each domain was examined only to the depth required by this translation problem. Foundational neuroscience was limited to memory encoding, hippocampal–neocortical

consolidation, sleep-dependent reactivation, sensory cue processing, and electrophysiological activity relevant to sleep monitoring. The technology analysis emphasized electroencephalography (EEG) and photoplethysmography (PPG), representing the principal electrophysiological and peripheral sensing pathways evaluated in this study. Actigraphy, cardiac measures, respiration, temperature, and other modalities were considered where they informed the interpretation of EEG-, PPG-, or multimodal systems.

A governing principle of the analysis was that measurement performance must be interpreted relative to its intended closed-loop function. Agreement with retrospective sleep staging does not by itself demonstrate that a sensing system can support causal state estimation, bounded decision latency, uncertainty handling, intervention control, or post-cue monitoring. General-purpose performance was therefore not treated as equivalent to intervention readiness.

The endpoint of the present study was an evidence-derived set of translational requirements and a justified first-generation implementation pathway. Detailed signal-processing development, machine-learning model training, software architecture, hardware integration, numerical latency thresholds, usability testing, longitudinal or clinical validation, regulatory analysis, and commercial evaluation were reserved for subsequent engineering and experimental work.

2.2 Evidence Identification and Source Selection

The study employed a structured, question-driven translational evidence synthesis. This design was selected because the research problem required the integration of heterogeneous forms of evidence serving different functions: mechanistic findings, experimental effects, measurement-validation results, real-time feasibility studies, and implementation constraints. The purpose was not to calculate a new pooled effect or exhaustively map a single literature, but to determine how evidence from several domains constrains a specific technology-development pathway.

The study should therefore not be interpreted as a formal systematic review, scoping review, or meta-analysis. Such review designs differ in purpose, retrieval requirements, appraisal procedures, and permissible conclusions ([Grant and Booth 2009](#); [Snyder 2019](#)). The present synthesis was organized around bounded research questions and prioritized evidential relevance and claim-to-decision traceability rather than exhaustive publication retrieval.

For each research question, sources were retained when they served at least one of the following functions:

- establishing a biological or physiological mechanism relevant to TMR;
- quantifying the efficacy, variability, or boundary conditions of TMR;
- validating a measurement method or wearable sensing modality;
- evaluating real-time or closed-loop operation;
- identifying a limitation affecting translation beyond the laboratory; or
- constraining or distinguishing a system-level implementation decision.

The evidence base included foundational neuroscience and sleep-medicine texts, peer-reviewed reviews and meta-analyses, primary experimental studies, independent

wearable-validation studies, and technical literature related to real-time physiological monitoring. Primary studies were preferred for specific experimental findings and device-performance claims. Reviews and meta-analyses were used to evaluate effect magnitude, consistency, heterogeneity, and broader evidentiary boundaries. Recent empirical studies were prioritized in rapidly developing fields such as wearable sensing, automated sleep analysis, and home-based intervention, while authoritative texts and widely cited reviews were retained for comparatively stable foundational concepts.

Manufacturer documentation was consulted only when needed to establish publicly reported sensor configurations, device functions, data-access capabilities, or integration provisions. Manufacturer-reported performance was not treated as independent evidence of measurement accuracy, TMR efficacy, or closed-loop readiness. Wherever possible, device-performance claims were based on peer-reviewed comparisons with polysomnography or another appropriate physiological reference.

Sources were excluded from the central synthesis when they lacked direct relevance to a research question, duplicated conclusions already supported by stronger evidence, relied primarily on promotional claims, or could not inform a scientific or translational decision. Wearable-validation findings were treated cautiously when their populations, recording conditions, output definitions, or temporal assumptions differed substantially from the intended closed-loop function.

Source identification, appraisal, and synthesis were conducted by a single reviewer. Procedural safeguards included the use of predefined research questions, explicit source-function criteria, preference for primary and independently validated evidence, conservative treatment of conflicting findings, and traceability between each retained source and the research question it informed. Each source was mapped within the supporting Neuro-TMR research repository to preserve the connection between literature, interpretation, translational requirement, and implementation decision.

These procedures reduced unrestricted source selection but did not eliminate reviewer-dependent judgment. Duplicate screening, formal risk-of-bias scoring, standardized certainty grading, and a new quantitative synthesis were not performed. Quantitative effect estimates were therefore taken from published meta-analyses or individual studies rather than recalculated across the assembled literature.

2.3 Evidence Appraisal and Maturity Classification

The selected evidence was evaluated according to its strength, relevance, and translational value rather than publication count alone. Appraisal considered six dimensions:

- **Methodological validity:** study design, participant characteristics, control conditions, measurement methods, statistical reporting, and outcome definitions;
- **Consistency and replication:** convergence across independent studies, laboratories, populations, and experimental paradigms;
- **Effect magnitude and uncertainty:** reported effect sizes, confidence intervals, interindividual variability, and the distinction between statistical and practical significance;
- **Physiological specificity:** how directly a measurement or intervention represented the biological process required by the proposed closed-loop function;

- **Ecological validity:** the extent to which findings obtained under controlled conditions remained applicable to wearable, home-based, or unsupervised operation; and
- **Translational relevance:** whether a finding could constrain a system requirement, distinguish candidate pathways, or define a future validation question.

These dimensions informed qualitative critical appraisal; they were not converted into a numerical quality score. A methodologically strong study could still have limited translational relevance if its measurement, timing, or operational conditions differed substantially from those required for closed-loop TMR. Conversely, a study with narrow scope could remain translationally important if it directly tested a critical system assumption.

Conflicting findings were examined in relation to differences in participant characteristics, memory task, cue modality, cue intensity, sleep stage, sensing technology, study duration, and outcome definition. Disagreement was not resolved through simple frequency counting. When evidence remained heterogeneous or insufficiently replicated, the conclusion was restricted to the conditions directly supported by the literature.

Evidence maturity was classified separately from the four translational levels defined in Section 2.1. The translational levels specify the type of claim being made, whereas the maturity categories indicate how strongly that claim is supported.

- **Established evidence:** findings supported by convergent experimental results, independent replication, authoritative reviews or meta-analyses, or validated measurement methods;
- **Preliminary evidence:** findings supported by a limited number of studies, early home-based demonstrations, narrow populations, short study durations, or incomplete independent replication; and
- **Unresolved evidence:** claims for which mechanism, measurement validity, implementation performance, or intervention efficacy remained insufficient to support a first-generation system assumption.

These categories were interpretive rather than formal statistical grades. Their purpose was to prevent preliminary feasibility from being represented with the same confidence as convergent experimental evidence.

The appraisal also maintained a distinction between measurement accuracy and functional suitability. A sleep-staging system may achieve acceptable retrospective classification performance while remaining unsuitable for real-time intervention because of delayed output, inadequate signal-quality assessment, uncertain confidence calibration, restricted access to raw data, or inability to detect post-cue physiological changes. Performance claims were therefore interpreted in relation to the decision the sensing system was expected to support.

The output of this stage was an evidence-bounded conclusion for each research question. Each conclusion documented the strongest supported interpretation, the conditions under which it remained valid, the principal sources of uncertainty, and any assumption requiring subsequent experimental validation.

2.4 Evidence-to-Engineering Translation

The central methodological unit linked each scientific conclusion to a prospective engineering decision through four operations:

1. **Evidence-bounded conclusion:** the strongest interpretation supported by the literature, together with its conditions and limitations;
2. **Functional implication:** the capability a practical system would require for that conclusion to remain operationally relevant;
3. **System-level requirement:** a testable requirement describing what the system must measure, detect, control, or prevent; and
4. **Validation question:** an unresolved assumption that must be examined during engineering or experimental development.

For example, the evidence indicates that TMR cueing should occur during physiologically appropriate sleep conditions while avoiding arousal and sleep disruption. The functional implication is that a practical system must distinguish eligible intervals from unstable or unsuitable periods. The corresponding system-level requirement is that the controller must use ongoing physiological information to authorize, withhold, or suspend cue delivery. The remaining validation question is whether a selected sensing and control pathway can support these decisions reliably during overnight operation.

This intermediate translation prevents biological findings from being converted directly into a specific device, algorithm, or numerical threshold. A physiological function may potentially be implemented through more than one sensing modality. Candidate technologies were therefore compared only after the required function had been defined.

Figure 1 summarizes the complete six-stage workflow.

The six stages were:

1. **Research-question formulation.** Each investigation began with a bounded question specifying the phenomenon, the evidence needed to evaluate it, and its relevance to the overall translation problem.
2. **Structured literature identification.** Sources were selected according to their ability to establish a mechanism, quantify an effect, validate a measurement method, identify a limitation, or constrain a translational decision.
3. **Critical evidence synthesis.** Findings were evaluated across studies to identify convergence, disagreement, applicability boundaries, methodological limitations, and unresolved uncertainty.
4. **Functional translation.** Each evidence-bounded conclusion was reformulated as a statement of what a practical closed-loop system would need to accomplish.
5. **Pathway comparison and selection.** Candidate implementation pathways were evaluated against the derived requirements according to physiological relevance, temporal suitability, evidence maturity, operational burden, and validation risk.
6. **Engineering handoff.** The selected pathway and unresolved assumptions were transferred into a future program of component development, integration, prospective testing, and iterative refinement.

The resulting requirement set covered seven areas:

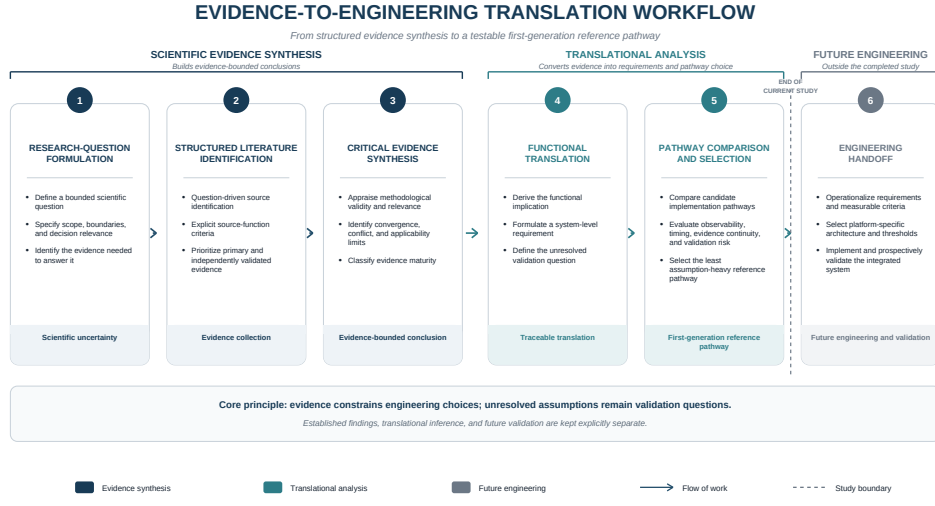


Fig. 1 Evidence-to-engineering translation workflow used in this study. Steps 1–3 formulate bounded research questions, identify relevant literature through a structured question-driven process, and produce evidence-bounded conclusions. Step 4 translates those conclusions into functional implications, system-level requirements, and unresolved validation questions. Step 5 compares candidate implementation pathways and selects the first-generation reference pathway according to physiological observability, evidentiary continuity, temporal suitability, and validation risk. Step 6 represents the subsequent engineering and prospective validation program and lies outside the completed scope of the present study. Unresolved assumptions are retained as validation questions rather than converted into predetermined engineering specifications. The figure is an original methodological synthesis by the author.

- **Physiological eligibility:** the states and conditions under which cueing may be appropriate;
- **Measurement quality:** the signal observability, reliability, physiological specificity, and uncertainty information required to support a decision;
- **Temporal validity:** the need for causal estimates, measurable processing delay, synchronized cue execution, and appropriately bounded intervention timing;
- **Intervention control:** the conditions under which cues must be delivered, withheld, modified, or terminated;
- **Disturbance prevention:** the detection and reduction of arousal, sleep disruption, inappropriate stimulation, or other unintended effects;
- **Operational reliability:** signal-quality monitoring, repeatability, fail-safe behavior, and robustness outside controlled laboratory conditions; and
- **Empirical validation:** the prospective tests required before an implementation assumption can be treated as a demonstrated capability.

Candidate pathways were evaluated according to how directly and reliably they could satisfy these functions. Commercial availability, comfort, or general-purpose sleep-staging performance were not sufficient grounds for selection when the corresponding closed-loop function remained unsupported. Where the evidence did not

permit a definitive translation, uncertainty was retained as a validation requirement rather than resolved through technological preference.

2.5 Study Outputs and Methodological Boundary

The methodology produced four outputs:

1. evidence-bounded conclusions concerning memory consolidation, TMR, wearable sensing, and closed-loop intervention;
2. an evidence-derived set of system-level translational requirements;
3. a structured comparison of candidate first-generation implementation pathways; and
4. selection of the pathway with the strongest continuity with the established evidence and the fewest unsupported assumptions.

These outputs define a scientifically justified starting point for subsequent development. They do not constitute evidence that an integrated system has been implemented, that its operating parameters have been determined, or that it produces reliable memory benefits outside the conditions represented in the reviewed literature.

The methodological conclusions are also bounded by the question-driven, single-reviewer design. The synthesis did not provide exhaustive literature retrieval, duplicate screening, formal risk-of-bias grading, or a new meta-analysis. These limitations reduce the certainty with which the resulting pathway can be generalized and are examined further in Section 7.

The present study therefore ends at pathway selection and qualitative requirement definition. Subsequent work must operationalize the requirements, select platform-specific architectures and thresholds, implement the closed-loop system, and evaluate its physiological reliability, behavioral efficacy, safety, and generalizability. The selected first-generation pathway should be interpreted as a testable development hypothesis rather than as a completed architecture or validated memory-enhancement product.

3 Neuroscientific Foundations of Memory and Sleep

3.1 Neural Systems Relevant to Memory and Sleep

The neurobiology relevant to Targeted Memory Reactivation (TMR) is distributed across interacting memory, sensory, and sleep-regulatory systems. For the present translation problem, the critical issue is not a complete account of human neuroanatomy, but the relationship among three functions: the encoding and reactivation of recently acquired memories, the regulation of sensory access during sleep, and the maintenance of physiological states in which consolidation can occur. These functions depend principally on hippocampal–neocortical, thalamocortical, and sleep–wake regulatory networks (Squire et al. 2015; Scammell et al. 2017).

The hippocampus and distributed neocortical networks occupy complementary roles in declarative memory. During learning, the hippocampus rapidly binds distributed features of an experience represented across cortical regions, allowing recently

acquired information to be retrieved as an integrated memory. Over time, reactivation of hippocampal–cortical representations is thought to contribute to systems-level reorganization and integration with existing cortical knowledge (Squire et al. 2015; Dudai et al. 2015). This process should not be interpreted as a simple transfer from a temporary hippocampal store to a permanent cortical store. The degree to which long-term memories become independent of the hippocampus varies across memory types and remains the subject of theoretical and empirical debate. For TMR, the important point is narrower: recent memories can remain susceptible to post-encoding modification through reactivation of distributed representations.

Sleep also changes how these memory systems interact with incoming sensory information. The thalamus participates in state-dependent regulation of thalamocortical communication and contributes directly to characteristic NREM oscillations, particularly sleep spindles. Sensory transmission is attenuated and reorganized during sleep rather than completely blocked, allowing the brain to reduce responsiveness to much of the external environment while retaining the capacity to process selected stimuli (Scammell et al. 2017; Fernandez and Lüthi 2020). This partial sensory accessibility is essential to the biological plausibility of TMR: an external cue must influence ongoing neural processing without producing sufficient activation to destabilize sleep.

Hypothalamic and brainstem circuits contribute to the regulation and stability of sleep–wake states, while structures including the amygdala can modulate the salience and persistence of particular memories. These systems are relevant to TMR primarily because they influence the physiological state in which cueing occurs and the probability that sensory stimulation produces arousal rather than memory-related processing. They are therefore treated here as modulatory components of a larger network rather than as independent memory-storage systems (Scammell et al. 2017).

3.2 Electrophysiological Basis of Brain-State Monitoring

For closed-loop TMR, the relevant bridge between neural physiology and engineering is population-level electrophysiology. Information is transmitted within and between neurons through electrical and chemical mechanisms, but scalp-level measurements do not resolve individual neuronal communication. Instead, coordinated activity across neuronal populations generates extracellular electrical fields whose temporal structure can be measured at progressively larger spatial scales.

Neural oscillations are recurring patterns of population activity produced by interactions within and between neuronal circuits. They span multiple frequencies and can organize the timing of neuronal excitability, communication, and plasticity (Buzsáki and Draguhn 2004). Frequency bands should not, however, be interpreted as one-to-one labels for specific cognitive states. Multiple rhythms coexist, their properties vary across brain regions and individuals, and their functional significance depends on the surrounding physiological context. During sleep, several oscillatory events become particularly informative because their occurrence and temporal relationships change systematically across sleep states.

Electroencephalography (EEG) provides a non-invasive measurement of voltage differences generated by these large-scale electrophysiological processes. Scalp EEG is influenced primarily by coordinated transmembrane currents in neuronal populations,

Figure 2. Functional Organization of Brain Systems Supporting Learning, Memory, Sensory Processing, and Sleep Regulation

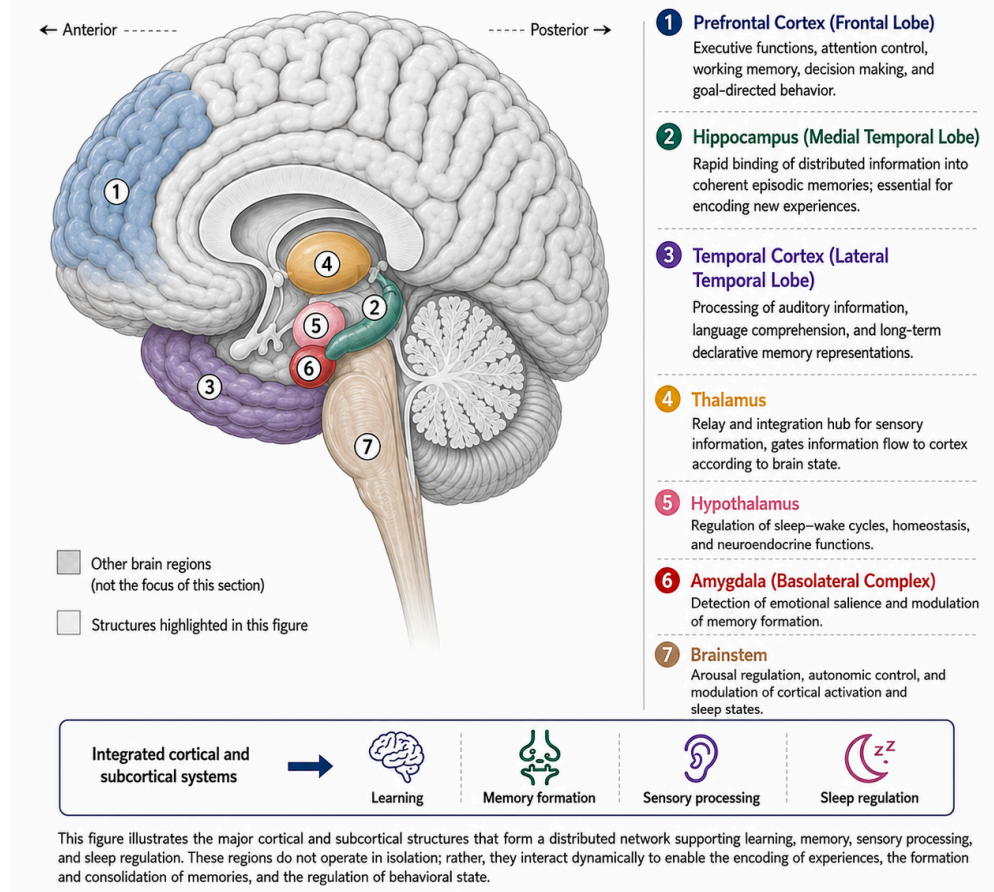


Fig. 2 Functional neural systems relevant to sleep-dependent memory processing. Hippocampal and distributed cortical networks support the encoding, reactivation, and longer-term reorganization of memory representations; thalamocortical circuits regulate state-dependent sensory processing and contribute to NREM oscillations; and hypothalamic and brainstem systems participate in sleep–wake and arousal regulation. The amygdala can additionally modulate memory according to emotional salience. The diagram represents a conceptual systems-level synthesis rather than anatomically independent functional modules (Squire et al. 2015; Scammell et al. 2017).

particularly synaptic currents in spatially organized cortical neurons, rather than by direct recording of individual action potentials (Buzsáki et al. 2012). EEG consequently provides high temporal resolution but limited spatial specificity: it can reveal when large-scale electrophysiological patterns occur without uniquely identifying the underlying neuronal sources.

This distinction is important when EEG is used for sleep monitoring. Conventional clinical sleep staging is based on polysomnography (PSG), which combines EEG

with electrooculography (EOG), electromyography (EMG), and other physiological channels. Standard scoring assigns wake, N1, N2, N3, and REM labels to successive epochs using characteristic combinations of these signals (Silber et al. 2007). EEG contributes strongly to this classification, particularly through changes in spectral activity and the appearance of events such as K-complexes, sleep spindles, and slow waves, whereas rapid eye movements are measured primarily by EOG and the reduction of skeletal-muscle tone characteristic of REM sleep is assessed using EMG.

Conventional sleep-stage labels and the underlying electrophysiological events also operate at different temporal scales. Manual staging is conventionally performed over 30-s epochs, whereas slow waves, spindles, arousals, and other transient events evolve over substantially shorter intervals (Silber et al. 2007). A real-time intervention may therefore require more than reproduction of conventional retrospective stage labels; it may also require access to the continuous signal from which intervention-relevant events and state transitions can be detected.

A further measurement boundary is particularly important for the later engineering analysis. Cortical slow waves and sleep spindles are observable in scalp EEG, whereas hippocampal sharp-wave ripples are primarily characterized through intracranial recordings and are not directly available from ordinary wearable scalp EEG. Human intracranial studies nevertheless show temporal coordination among hippocampal ripples, sleep spindles, and slow oscillations (Staresina et al. 2015). Accordingly, hippocampal ripple activity can inform the biological model of sleep-dependent consolidation without being treated as a directly measurable control signal in a first-generation wearable EEG system.

3.3 Memory Consolidation and Reactivation

Encoding does not immediately produce a fixed long-term memory. Newly acquired representations remain susceptible to interference and modification, and their persistence depends on biological processes collectively described as memory consolidation. Consolidation operates across multiple temporal and spatial scales, from cellular and synaptic modifications within initially engaged circuits to systems-level changes in how memories are represented across hippocampal and cortical networks (Squire et al. 2015; Dudai et al. 2015).

Synaptic consolidation refers broadly to post-learning cellular processes that stabilize experience-dependent changes within local circuits. Systems consolidation concerns longer-term reorganization of distributed memory representations and changes in interactions between the hippocampus and neocortex. These processes should not be treated as a single linear transfer mechanism: their time courses overlap, their expression depends on the type of memory, and multiple theoretical accounts differ in the extent to which remote memories continue to require hippocampal involvement (Squire et al. 2015; Dudai et al. 2015).

A central process linking learning to later systems-level change is memory reactivation. Neural patterns related to previous experience can reappear after encoding during both wakefulness and sleep. During NREM sleep, such reactivation is frequently observed in temporal association with hippocampal sharp-wave ripples and with cortical and thalamocortical oscillations implicated in memory processing (Rasch and Born

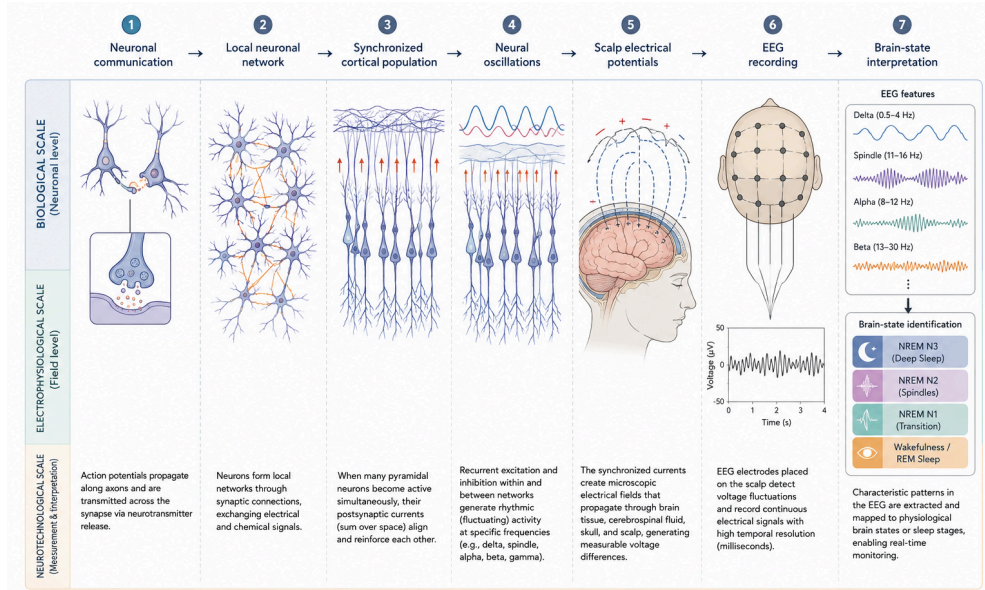


Figure 3. From neurons to EEG and brain-state interpretation. Neuronal communication at the cellular level scales up to synchronized activity in large cortical populations. The resulting oscillatory electrical fields propagate to the scalp, where they are detected by EEG electrodes. The recorded signals contain time-resolved information that can be used to infer the current brain state, including different sleep stages.

Fig. 3 Conceptual relationship between neuronal population activity and EEG-based brain-state interpretation. Coordinated transmembrane currents across neuronal populations generate extracellular fields that contribute to measurable scalp potentials. Their temporal organization produces EEG features, including oscillatory activity and transient events, that can be used as physiological indicators of sleep state. Scalp EEG reflects population-level activity and does not directly record individual neurons or deep hippocampal sharp-wave ripples (Buzsáki et al. 2012).

2013; Klinzing et al. 2019). Reactivation is therefore widely considered an important component of contemporary models of sleep-dependent systems consolidation.

The mechanistic interpretation nevertheless requires caution. Reactivation is not synonymous with consolidation, and the occurrence of replay-like activity does not imply that every reactivation event strengthens a memory. Instead, reactivation provides an opportunity for previously encoded representations to be modified, integrated, or stabilized under physiological conditions that support plasticity. The behavioral consequences depend on the memory, surrounding neural state, and temporal organization of the relevant activity (Dudai et al. 2015; Klinzing et al. 2019).

This distinction is central to TMR. A sensory cue presented during sleep cannot be assumed to create or restore a memory that was not successfully encoded. Rather, TMR depends on an association established during wakefulness and seeks to bias subsequent reactivation toward the corresponding representation. The scientific target of the intervention is therefore an endogenous post-encoding process, not memory formation independently of prior learning.

3.4 Sleep Architecture and Oscillatory Coordination of Memory

Human sleep is organized into recurring transitions among NREM and REM states. Under contemporary clinical scoring conventions, NREM sleep is divided into N1, N2, and N3. N1 represents the transition from wakefulness into sleep; N2 is characterized in part by K-complexes and sleep spindles; and N3 contains prominent high-amplitude slow-wave activity and is commonly referred to as slow-wave sleep (SWS) in human experimental research. REM sleep differs from NREM sleep through a combination of relatively low-amplitude mixed-frequency EEG activity, rapid eye movements, and reduced skeletal-muscle tone (Silber et al. 2007).

These stages should not be interpreted as discrete functional compartments in which a single cognitive process occurs exclusively. Memory-related processing is distributed across the sleep period, and different forms of learning can show different relationships with NREM and REM physiology (Diekelmann and Born 2010; Rasch and Born 2013). For the declarative memory processes most relevant to much of the TMR literature, NREM sleep—particularly N2 and N3—provides a well-supported physiological context for examining reactivation and hippocampal–neocortical communication. This does not imply that N3 is universally optimal for every memory domain or cueing protocol.

A prominent mechanistic framework for NREM-dependent systems consolidation centers on the temporal coordination of cortical slow oscillations, thalamocortical sleep spindles, and hippocampal sharp-wave ripples. Slow oscillations organize alternating periods of cortical excitability; spindles emerge from thalamocortical circuitry and are associated with transient windows of cortical plasticity; and hippocampal sharp-wave ripples are closely associated with replay of recently experienced activity. Converging evidence suggests that their temporal nesting can support communication between hippocampal and cortical memory systems (Staresina et al. 2015; Fernandez and Lüthi 2020; Klinzing et al. 2019).

The coupling of these events is scientifically important but should not be converted prematurely into a requirement for phase-specific stimulation. Evidence that oscillatory coordination contributes to consolidation does not, by itself, establish that a first-generation TMR system must detect every event or deliver cues at a particular oscillatory phase. Stage-aware cueing and event- or phase-aware cueing therefore represent distinct levels of intervention control. Whether the additional temporal precision of the latter produces a reproducible behavioral advantage is an empirical question addressed later in the translational analysis.

For wearable implementation, this distinction also separates the biological mechanism from what can actually be observed. Scalp EEG can provide information about sleep stage, cortical slow-wave activity, and sleep spindles, while deeper hippocampal processes remain indirectly inferred. The relevant engineering question is consequently not whether every component of the consolidation mechanism can be measured non-invasively, but whether the observable signals provide sufficient information to identify physiologically appropriate conditions for intervention.

Together, these findings establish the biological constraints relevant to TMR: a previously encoded representation must remain capable of reactivation, cueing must

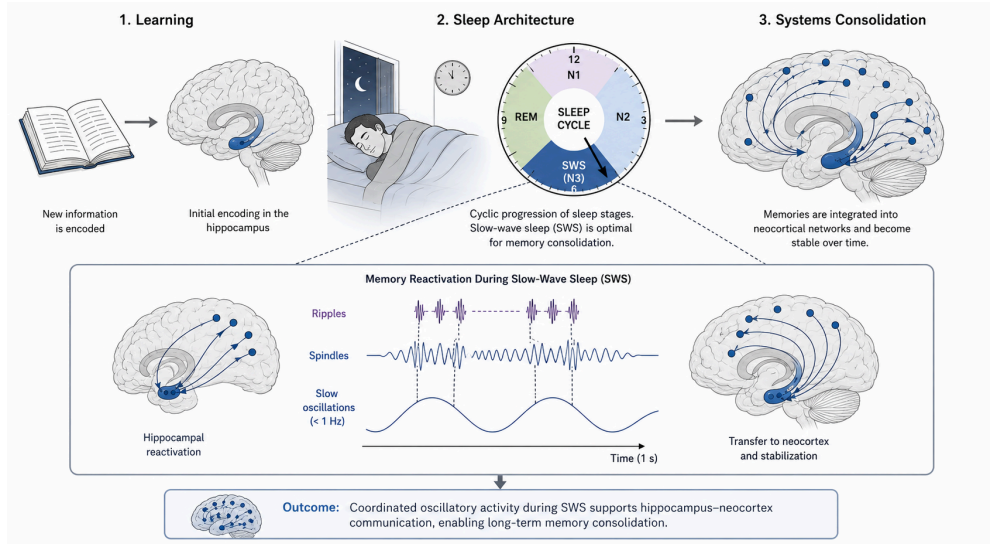


Fig. 4 Conceptual framework of sleep-dependent systems memory consolidation. Recently encoded memories initially depend strongly on interactions between hippocampal and distributed cortical representations. During NREM sleep, memory reactivation occurs within a physiological environment characterized by coordinated cortical slow oscillations, thalamocortical sleep spindles, and hippocampal sharp-wave ripples. Their temporal interaction is proposed to support communication and longer-term reorganization across memory networks. The figure represents a mechanistic synthesis rather than a claim that consolidation proceeds through a single deterministic pathway (Rasch and Born 2013; Staresina et al. 2015; Klinzing et al. 2019).

occur within an appropriate physiological state, and the intervention must preserve rather than disrupt the sleep processes associated with consolidation. Section 4 examines the extent to which experimental TMR can exploit these conditions to produce measurable changes in subsequent memory.

4 Targeted Memory Reactivation: From Biological Mechanism to Closed-Loop Intervention

4.1 Experimental Principle of Targeted Memory Reactivation

Targeted Memory Reactivation (TMR) is an experimental procedure for biasing post-encoding memory processing through sensory cues associated with prior learning. During wakefulness, a sound, odor, or other stimulus is paired with specific learned information. Re-representation of that cue during subsequent sleep can selectively influence later retention of the associated material (Rasch et al. 2007; Rudoy et al. 2009; Oudiette and Paller 2013). TMR therefore does not introduce information into memory during sleep; it acts on representations established before sleep by providing an external signal linked to the earlier learning episode.

The mechanistic interpretation of this effect requires caution. A selective behavioral benefit following cueing is evidence that the cue altered post-encoding memory

processing, but it does not by itself demonstrate the precise neural content or pathway of reactivation. Neuroimaging and electrophysiological studies provide additional evidence that sleep-time cues can evoke learning-related activity and content-specific neural patterns (Rasch et al. 2007; Cairney et al. 2018; Wang et al. 2019). The most defensible interpretation is therefore that appropriately associated cues can bias endogenous memory reactivation during sleep, while the exact neural sequence linking cue perception to later behavioral improvement remains incompletely resolved.

This distinction separates TMR from nonspecific sensory stimulation. Its experimental logic depends on three linked conditions: the existence of a pre-sleep cue–memory association, presentation of the cue under an appropriate physiological condition, and a subsequent difference in memory attributable to the cueing manipulation. These conditions form the basis for evaluating both TMR efficacy and its later translation into an automated intervention.

4.2 Encoding and Cue–Memory Association

The effectiveness of TMR is constrained by what has been learned before sleep. A cue can only influence a memory insofar as it has acquired a meaningful relationship with the target representation during encoding. Evidence from object–location and associative-learning paradigms indicates that prior memory strength and the directness of the cue–memory association can influence the magnitude of subsequent TMR effects (Creery et al. 2015; Cairney et al. 2016).

This relationship is more nuanced than a simple requirement for maximally strong encoding. Creery et al. found that near-perfectly learned spatial memories did not show the same TMR benefit as less completely learned memories, while Cairney et al. reported stronger effects for memories recalled with relatively low pre-sleep accuracy and found evidence that direct cue–memory associations were important for successful reactivation (Creery et al. 2015; Cairney et al. 2016). TMR should therefore not be described as rescuing memories that were never encoded, but neither should weak pre-sleep performance automatically be treated as a failure condition.

Cue specificity also affects interpretation. Item-specific auditory cues can permit selective comparison between cued and uncued memories, as demonstrated by Rudoy et al., whereas contextual cues such as odors can reactivate a broader learning context (Rasch et al. 2007; Rudoy et al. 2009). Behavioral selectivity should not be equated with complete neural isolation: a cue may reactivate features of the target memory together with contextual or associated representations. For translational purposes, the critical requirement is therefore not maximal cue uniqueness in the abstract, but a reproducible association between the cue and the memory process the intervention is intended to influence.

4.3 Physiological Conditions for State-Dependent Cueing

TMR efficacy depends not only on what is cued, but also on the physiological state in which cueing occurs. The meta-analysis by Hu et al. found significant positive effects for cueing during NREM stage 2 ($g = 0.32$, 95% CI 0.04–0.60) and slow-wave sleep ($g = 0.27$, 95% CI 0.20–0.35), whereas pooled effects for REM sleep and wakefulness

were not significant (Hu et al. 2020). These results support NREM sleep as the best-established cueing context in the literature synthesized by that analysis, but they do not establish that a single sleep stage is universally optimal for every memory domain or experimental objective.

The physiological context described in Section 3 provides a plausible basis for this state dependence. During NREM sleep, slow oscillations, sleep spindles, and hippocampal activity associated with memory reprocessing exhibit structured temporal relationships (Rasch and Born 2013; Klinzing et al. 2019; Staresina et al. 2015). TMR cues enter this ongoing physiological environment rather than initiating consolidation independently. Accordingly, cue timing should be understood as an interaction with endogenous sleep dynamics rather than as delivery of a stimulus to a uniformly receptive sleeping brain.

Evidence also indicates that finer temporal structure within NREM can influence cue efficacy. Cue-induced spindle activity has been associated with successful memory processing, and manipulation of cue timing relative to spindle dynamics has demonstrated that some NREM intervals are more favorable than others (Cairney et al. 2018; Antony et al. 2018). More direct closed-loop evidence has subsequently tested TMR relative to ongoing slow-oscillation phase. Shimizu et al. implemented an EEG-based closed-loop system that delivered memory-associated sounds around detected slow-oscillation down-to-up transitions and reported improved spatial-navigation performance relative to a non-stimulated control condition (Shimizu et al. 2018). The study demonstrated the feasibility and behavioral relevance of phase-sensitive closed-loop TMR, although it did not establish superiority over an otherwise equivalent open-loop TMR protocol.

Ngo and Staresina later directly compared memory cues delivered during slow-oscillation UP and DOWN states and found reduced overnight forgetting for memories cued during UP states relative to DOWN states (Ngo and Staresina 2022). More recently, Nicolas et al. extended this approach to motor memory and likewise found phase-dependent behavioral and electrophysiological effects, with greater overnight performance improvement for UP- than DOWN-reactivated motor sequences (Nicolas et al. 2025). Importantly, the UP-reactivated condition in that study did not significantly outperform the non-reactivated condition, illustrating that evidence for phase dependence does not yet establish a general incremental advantage of phase-locked TMR over simpler cueing strategies.

These findings provide direct empirical support for event- and phase-aware closed-loop TMR rather than merely a mechanistic rationale for it. The evidence base is nevertheless narrower than that supporting TMR delivered within broader NREM conditions, and phase-specific control introduces additional requirements for causal event detection, phase estimation, synchronization, and cue timing. Stage-aware and phase-aware control should therefore remain distinct levels of intervention complexity.

These findings provide the rationale for state-dependent closed-loop control in an autonomous implementation. TMR itself does not require an automated closed-loop system; many foundational studies relied on experimenter-supervised sleep monitoring. For real-world automation, however, physiological information must replace continuous researcher judgment. Such a system must determine when cueing is permitted,

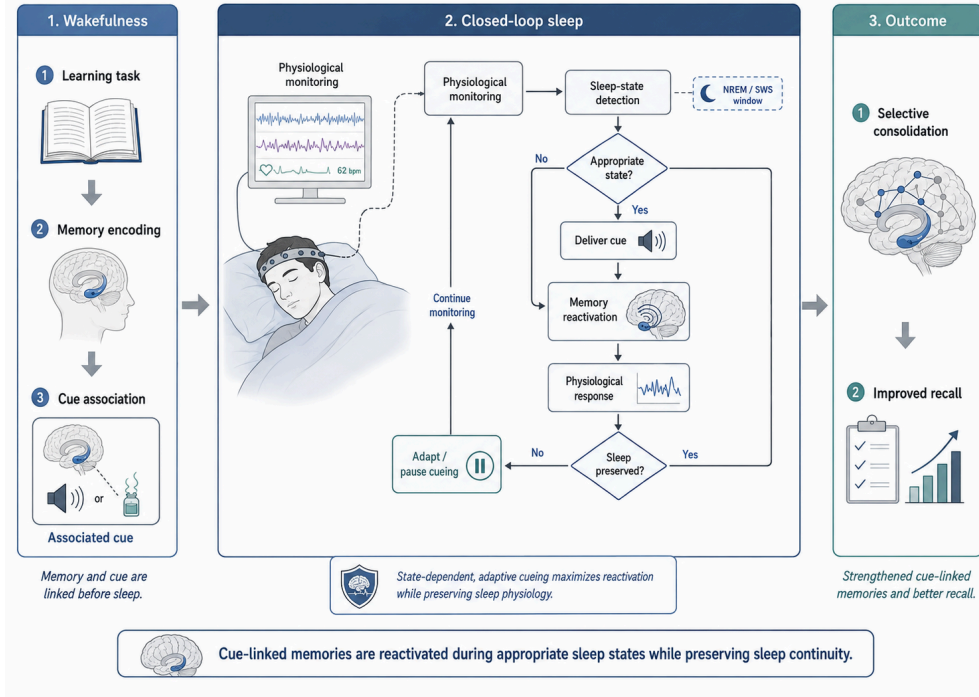


Fig. 5 Conceptual workflow for state-dependent closed-loop Targeted Memory Reactivation. Sensory cues are first associated with selected information during wakeful learning. During subsequent sleep, ongoing physiological measurements are used to determine whether the current state is eligible for cue presentation. Following stimulation, physiological information can be used to evaluate disturbance and determine whether cueing should continue, be delayed, or be suspended. The workflow is an original conceptual synthesis of experimental and automated TMR paradigms; cue delivery does not guarantee memory enhancement, and behavioral efficacy remains an empirical outcome (Oudiette and Paller 2013; Whitmore et al. 2022; Salfi et al. 2025).

when it should be withheld, and when evidence of disturbance should interrupt further stimulation. The quantitative thresholds and algorithms needed to implement these functions remain engineering and validation questions rather than established neuroscientific constants.

4.4 Experimental Evidence for TMR Efficacy and Mechanism

Early controlled experiments established the fundamental causal effect of sleep-time cue presentation on later memory performance. Rasch et al. re-presented an odor that had been present during learning and found improved retention of a hippocampus-dependent spatial task when the odor was presented during slow-wave sleep. The same manipulation was ineffective when the odor had not been associated with learning and did not produce the corresponding benefit during wakefulness or REM sleep (Rasch et al. 2007). The experiment therefore demonstrated that the behavioral effect depended jointly on prior association and physiological state rather than on sensory stimulation alone.

Rudoy et al. subsequently demonstrated item-level selectivity using an object–location task. Individual objects were paired with characteristic sounds during learning, and only a subset of those sounds was replayed during sleep. Post-sleep location memory was more accurate for cued than uncued objects (Rudoy et al. 2009). Because cued and uncued items were learned by the same individuals and experienced the same sleep period, this within-subject design provided strong evidence that targeted sensory reminders can alter the retention of selected memories.

The broader magnitude of the effect was quantified by Hu et al. in a multilevel meta-analysis of 91 experiments, 212 effect sizes, and 2,004 participants. Across sleep-based TMR experiments, the pooled effect was Hedges’ $g = 0.29$ (95% CI 0.21–0.38) (Hu et al. 2020). This represents a modest standardized benefit, not a 29% improvement in memory performance and not an expected effect for every participant or protocol.

The same meta-analysis also places an important boundary around that estimate. Effect sizes were substantially heterogeneous ($I^2 = 71\%$), indicating large variation beyond sampling error. Hu et al. also detected funnel-plot asymmetry consistent with possible publication bias; under one trim-and-fill correction, the estimated overall effect decreased to $g = 0.18$ (95% CI 0.06–0.30) while remaining significantly above zero (Hu et al. 2020). The central quantitative conclusion is therefore not that TMR produces a fixed effect of $g = 0.29$, but that the literature supports a small positive average effect whose magnitude is strongly condition-dependent.

Neurophysiological findings provide convergent but not definitive evidence about the mechanism underlying this behavioral effect. Cairney et al. found that memory cues increased fast-spindle activity relative to control stimuli and that memory-category information could be decoded during this spindle response; the fidelity of decoding predicted later TMR benefit (Cairney et al. 2018). Wang et al. likewise identified content-related EEG activity following sleep-time cues, with stronger learning-related signals associated with subsequent memory and post-cue spindle activity (Wang et al. 2019). Antony et al. further showed that experimentally manipulating cue timing relative to ongoing spindle dynamics changed later memory performance (Antony et al. 2018).

These studies strengthen the interpretation that TMR can engage memory-specific processing embedded within NREM electrophysiology. They do not, however, justify treating any single EEG response as a validated surrogate for successful memory enhancement. Spindle activity and decodable neural patterns are mechanistically informative, but their relationship to behavioral benefit remains variable and in several studies correlational. Behavioral outcome therefore remains necessary when evaluating TMR efficacy.

4.5 From Laboratory Demonstration to Home-Based Feasibility

Moving TMR beyond supervised laboratory environments introduces a different evidentiary question: whether the intervention can preserve its physiological and behavioral properties when monitoring and cue delivery are automated. Early unsupervised work demonstrated the difficulty of this transition. Göldi and Rasch applied

vocabulary cueing across three nights at home without online sleep-state control and found no uniform benefit across the full sample; outcomes depended strongly on reported sleep disturbance and habituation (Göldi and Rasch 2019). This result is important because it shows that removing physiological supervision does not automatically preserve laboratory efficacy.

Whitmore et al. subsequently developed SleepStim, which combined smartwatch movement and heart-rate measurements with a smartphone-based auditory cueing system. In the first experiment ($n = 61$), the overall TMR effect was not significant, although memory outcome varied systematically with sound intensity. After restricting stimulus intensity in a second experiment ($n = 24$), cued spatial memories showed a significant advantage (Whitmore et al. 2022). The study provides evidence that automated peripheral-sensor-guided TMR can operate at home while simultaneously demonstrating that intervention parameters and sleep disturbance materially affect the result.

More direct electrophysiological control was demonstrated by Salfi et al. using a Dreem 2 wearable EEG system in 24 adults sleeping at home. Participants learned translations for 40 pseudowords, half of which were subsequently re-presented following real-time detection of slow waves. Cued items improved by 8.6% from evening test to morning retest, whereas the corresponding change for uncued items was not significant; morning recall was also higher for cued than uncued material. Successful recall was associated with greater post-stimulus spindle-range activity (Salfi et al. 2025).

Taken together, these studies provide proof-of-concept evidence for automated home TMR across both peripheral and EEG-based sensing approaches (Göldi and Rasch 2019; Whitmore et al. 2022; Salfi et al. 2025). They do not establish population-level effectiveness, long-term reliability, or equivalence between sensing modalities. The evidence instead demonstrates a progression from unsupervised cue playback to physiological state-dependent automation, while showing that sensing quality, stimulation control, and preservation of sleep remain central determinants of translational success.

4.6 Boundary Conditions and Evidence-Derived Implications

TMR should therefore be interpreted as a condition-dependent intervention rather than a deterministic memory-enhancement mechanism. Its pooled behavioral effect is positive but modest, and substantial between-study heterogeneity indicates that protocol, memory type, sleep state, and participant-level factors materially influence outcome (Hu et al. 2020). Prior memory strength and the structure of cue-memory associations provide additional sources of variability (Creery et al. 2015; Cairney et al. 2016).

Individual predictability remains particularly limited. Differences in sleep architecture, sensory threshold, initial learning, cue responsiveness, and oscillatory physiology may contribute to variation between participants and nights, but currently available neural markers should not be treated as prospectively validated predictors of individual benefit. A strong cue-evoked spindle response can be associated with successful memory processing without guaranteeing that a specific memory will subsequently improve.

The durability and generalization of TMR effects also remain incompletely characterized. Much of the evidence relevant to the present synthesis measures memory over relatively short post-sleep intervals, while only limited work has examined repeated unsupervised use across multiple nights (Göldi and Rasch 2019; Whitmore et al. 2022). Evidence that TMR improves a specific cued memory should therefore not be generalized to durable cognitive enhancement, broad learning improvement, or cumulative benefits from indefinite repeated stimulation.

Mechanistic uncertainty likewise remains relevant to intervention design. Cue-evoked spindle activity, content-specific neural decoding, and temporal sensitivity to oscillatory state support contemporary reactivation models (Cairney et al. 2018; Wang et al. 2019; Antony et al. 2018), but they do not yet establish a single optimal triggering rule. In particular, evidence that fine-scale oscillatory timing influences TMR should not be converted directly into a requirement for phase-specific first-generation control. Whether event-aware or phase-aware cueing provides a reproducible advantage over well-controlled stage-aware TMR remains an empirical comparison.

The evidence reviewed in this section nevertheless establishes several translational constraints. A practical TMR system must operate on memories that have been associated with recoverable cues before sleep; cueing must be limited to physiologically supported sleep conditions; stimulation must be controlled to reduce sleep disturbance; and intervention timing and physiological context must be recorded so that behavioral outcomes can be interpreted in relation to actual system operation. These conclusions define the biological requirements of the intervention without determining which sensing technology can satisfy them. Section 5 therefore evaluates the wearable measurement pathways available for observing those conditions in real time.

5 Wearable Sleep Monitoring

5.1 Polysomnography and the Physiological Measurement Target

Polysomnography (PSG) provides the reference framework for characterizing human sleep because sleep stages are defined from multiple physiological signals rather than from behavior or movement alone. Standard PSG combines electroencephalography (EEG), electrooculography (EOG), and electromyography (EMG), with additional respiratory and cardiovascular channels included when required. Under conventional scoring rules, these signals are interpreted together to classify successive 30-s epochs as wake, N1, N2, N3, or rapid eye movement (REM) sleep (Silber et al. 2007).

For TMR, the relevance of PSG extends beyond the categorical stage label. EEG also preserves information about transient electrophysiological events such as slow waves, sleep spindles, K-complexes, and cortical arousals. These events can occur at substantially shorter timescales than the epochs used for conventional sleep staging. A recording may therefore be correctly classified as N2 or N3 while still containing moment-to-moment physiological variation relevant to whether stimulation should occur.

PSG should nevertheless be treated as a standardized reference rather than as an error-free representation of a continuously evolving brain state. Human sleep scoring contains genuine ambiguity, particularly near stage transitions. Rosenberg and Van Hout reported an average inter-scorer stage agreement of 82.6% in the American Academy of Sleep Medicine inter-scorer program, with lower agreement for N3 and especially N1 than for several other stages (Rosenberg and Van Hout 2013). Wearable performance should consequently be interpreted relative to a high-quality physiological reference whose own categorical labels contain uncertainty.

The translational objective is therefore not to reproduce the complete PSG laboratory in wearable form. It is to determine which components of the physiological information available in PSG must remain observable for a particular closed-loop decision. This distinction permits reduced sensing systems to be evaluated according to functional adequacy rather than according to whether they reproduce every channel used in clinical polysomnography.

5.2 Wearable EEG: Preserving Electrophysiological Observability

Wearable EEG attempts to preserve access to scalp electrophysiology while reducing the recording burden of full PSG. Compared with peripheral sensing, its principal scientific advantage is not that it measures sleep “directly,” but that the recorded signal contains electrophysiological features from which sleep stages and several intervention-relevant events are conventionally defined. Depending on montage and signal quality, wearable EEG may preserve information about slow-wave activity, sleep spindles, stage transitions, and cortical responses to sensory stimulation.

Reducing the number of electrodes introduces important trade-offs. Fewer channels reduce spatial coverage and redundancy and may make it more difficult to distinguish localized artifact from genuine physiology. They do not, however, inherently remove EEG’s high temporal resolution. The relevant question is whether a particular montage preserves the signal features required by the intended decision and whether those features remain reliable throughout overnight use.

Empirical validation demonstrates both the potential and the limitations of this approach. In a developer-associated validation of the Dreem headband, Arnal et al. compared simultaneous headband and PSG recordings in 25 participants. Automatic sleep staging achieved a mean overall accuracy of $83.5 \pm 6.4\%$, compared with $86.4 \pm 8.0\%$ average agreement for five human sleep experts in the same study (Arnal et al. 2020). These results showed that reduced-montage wearable EEG can preserve substantial sleep-staging information under controlled recording conditions.

Performance should not, however, be generalized from a single device, population, or validation design. An independent evaluation of the same headband in 62 older adults, including participants with Alzheimer’s disease, reported moderate five-stage epoch-level performance ($MCC = 0.54 \pm 0.14$); after artifact removal, 73% of recordings were usable for quantitative EEG analysis (Ravindran et al. 2025). The contrast between these studies illustrates that wearable EEG performance depends on population, signal quality, recording conditions, algorithm behavior, and the physiological quantity being extracted.

This evidence supports wearable EEG as a plausible means of retaining electrophysiological observability outside full PSG, but not as an automatically reliable solution. For closed-loop TMR, stage classification must therefore be evaluated together with signal-quality detection, availability of continuous data, preservation of intervention-relevant EEG features, and the ability to withhold interpretation during corrupted intervals.

5.3 PPG and Peripheral Sleep-State Inference

Photoplethysmography (PPG) approaches sleep monitoring from a different physiological level. PPG measures peripheral blood-volume changes associated with the cardiac cycle, from which pulse rate, pulse-rate variability, waveform characteristics, and respiratory-related information can be derived. Because autonomic regulation changes across wakefulness, NREM sleep, and REM sleep, these signals contain information that can be used to estimate sleep state probabilistically (de Zambotti et al. 2019).

The distinction from EEG is therefore one of observability rather than simple accuracy. PPG does not observe the cortical electrical phenomena used to define slow waves, sleep spindles, or cortical arousals. Instead, an algorithm infers sleep state from peripheral physiological correlates. Actigraphy, temperature, respiration, electrodermal activity, and other peripheral channels can be added to reduce ambiguity, but the resulting estimate remains an inference from signals downstream from or correlated with the neural state being estimated.

This inferential distance does not imply that PPG-based sleep staging is ineffective. Korkalainen et al. trained deep-learning models using PPG recordings from 894 patients undergoing diagnostic PSG. In an independent test set, three-stage classification of wake, NREM, and REM reached 80.1% accuracy with Cohen’s $\kappa = 0.65$. Performance declined as the classification problem became more specific: four-stage classification reached 68.5% ($\kappa = 0.54$), while five-stage classification reached 64.1% ($\kappa = 0.51$) (Korkalainen et al. 2020). These results demonstrate that peripheral cardiac information contains substantial sleep-state information while also showing the cost of resolving more specific physiological categories.

Consumer and multisensor devices show similarly variable results. Chinoy et al. compared seven consumer sleep-tracking systems with simultaneous PSG in healthy young adults. Sleep sensitivity was high across devices (≥ 0.93), but wake specificity ranged from 0.18 to 0.54, sleep-stage performance was inconsistent, and several devices performed worse during disrupted sleep (Chinoy et al. 2021). A more recent multi-night study of the Oura Ring Generation 3 reported strong agreement with PSG for several global sleep measures and stage-specific classification, although the study received financial support from the device manufacturer and should be interpreted in that context (Svensson et al. 2024). Together, these studies show that peripheral sensing can provide useful and sometimes strong retrospective sleep estimates, but performance is device-, algorithm-, population-, and condition-dependent.

Importantly, peripheral sensing has already progressed beyond retrospective sleep summaries in TMR research. SleepStim used smartwatch-derived heart-rate and inertial signals to estimate the probability of N3 sleep in real time and to govern auditory

Table 1 Comparison of sleep-monitoring modalities relevant to closed-loop TMR. The table distinguishes the physiological information observed by each modality from the inference or intervention decision subsequently made from that information.

Modality	Physiological information	Representative evidence	TMR-relevant limitation
PSG	Multimodal EEG, EOG, and EMG, with additional physiological channels as required.	Reference framework for standardized sleep staging and observation of stage-specific electrophysiology (Silber et al. 2007).	High measurement burden; conventional stage labels remain epoch-based and contain scorer uncertainty.
Wearable EEG	Reduced-montage scalp electrophysiology with device-dependent access to slow-wave, spindle, arousal, and stage-related features.	Dreem validation reported 83.5% mean staging accuracy in one controlled study; performance was more moderate in an independent older population (Arnal et al. 2020; Ravindran et al. 2025).	Reduced spatial redundancy and vulnerability to overnight artifact and electrode-contact loss; performance is configuration- and population-specific.
PPG / cardiac	Peripheral pulse and autonomic dynamics associated with sleep-state changes.	PPG-only models have demonstrated moderate-to-substantial agreement with PSG, with performance declining as stage specificity increases (Korkalainen et al. 2020).	Does not directly observe cortical slow waves, spindles, or cortical arousal; stage must be inferred from peripheral correlates.
Multimodal peripheral	PPG combined with movement and potentially temperature, respiration, or other peripheral channels.	Consumer systems can provide useful sleep estimates, but performance varies substantially across devices and recording conditions (Chinoy et al. 2021; Svensson et al. 2024).	Additional signals may improve stage inference without establishing electrophysiological equivalence or intervention readiness.
Actigraphy	Movement and sustained immobility measured through accelerometry.	Well suited to longitudinal sleep-wake estimation but provides limited physiological stage information (de Zambotti et al. 2019).	Insufficient alone for distinguishing the electrophysiologically defined NREM states required by stage-aware TMR.

Note. Performance values are not intrinsic properties of sensor modalities. They depend on device configuration, preprocessing, classification algorithm, population, recording environment, reference scoring procedure, and validation design.

cue delivery at home (Whitmore et al. 2022). This constitutes direct proof of concept that peripheral physiological information can participate in automated TMR control. It does not establish equivalence to EEG-based control, nor does it demonstrate that general consumer sleep-stage outputs are automatically suitable for TMR. Instead, it shows that the scientific question must be framed at the level of the complete decision process rather than sensor type alone.

5.4 Validation Evidence and What Accuracy Means

Validation of wearable sleep monitoring generally relies on simultaneous recording with PSG followed by epoch-level or summary-level comparison. Such a comparison is necessary but not sufficient for interpretation. The apparent performance of a system depends on the definition of the classification task, the composition of the validation

population, the independence of the test data, the quality of the reference scoring, and the metrics used to summarize error (de Zambotti et al. 2019).

The number of predicted classes is particularly important. Sleep–wake classification, three-stage classification, four-stage classification, and full five-stage scoring are progressively different problems. Results obtained for one should not be presented as evidence for another. The PPG results reported by Korkalainen et al., for example, decreased from 80.1% accuracy for three-stage classification to 64.1% for five-stage classification (Korkalainen et al. 2020). A single headline accuracy value therefore has little meaning unless the target classes are specified.

Overall accuracy can also conceal stage-specific failure. Sensitivity indicates how often a true target state is recognized; precision indicates how often a predicted target state is actually correct; specificity quantifies rejection of non-target states; and confusion matrices reveal which physiological states are systematically mistaken for one another. Cohen’s kappa, Matthews correlation coefficient, macro-F1, and related measures can reduce some distortions caused by class imbalance, although no single metric captures every operational consequence.

For TMR, false-negative and false-positive errors have different implications. Failing to recognize one eligible NREM interval may simply remove an opportunity for stimulation. Incorrectly declaring an unsuitable interval eligible may cause a cue to be delivered outside the physiological conditions supported by the evidence. The relative importance of sensitivity and precision must therefore be determined from the intervention policy rather than from generic sleep-staging conventions.

Generalization is equally important. Consumer-device studies show that performance can deteriorate during disrupted sleep and vary across algorithms and devices (Chinoy et al. 2021). Wearable EEG evidence likewise changes across populations and signal-quality conditions (Arnal et al. 2020; Ravindran et al. 2025). Validation results should therefore be interpreted as properties of a particular sensor–algorithm–population combination, not as universal properties of “EEG” or “PPG.”

Independence of validation also matters. Developer- or manufacturer-supported studies can provide valuable peer-reviewed evidence, but their provenance should remain visible and their results should be considered together with independent validation. The purpose of this distinction is not to discard industry-linked evidence, but to avoid treating a single favorable validation as sufficient for a modality-level conclusion.

5.5 From Retrospective Scoring to Real-Time Closed-Loop Control

Retrospective sleep scoring and real-time intervention control solve different problems. An offline algorithm may use complete-night context, future epochs, temporal smoothing, repeated preprocessing, and manual artifact review before assigning the final hypnogram. A causal controller has access only to data already acquired and must make a decision before the physiological opportunity for stimulation has passed.

This distinction does not imply that all non-automated TMR protocols are scientifically invalid. Foundational TMR experiments often relied on experimenters who monitored sleep physiology and manually controlled stimulation. The limitation arises

when an autonomous system uses a fixed or unsupervised schedule without verifying that the sleeper remains in an eligible state. For such a system, physiological state must influence whether cue delivery proceeds, is delayed, or is withheld.

Latency becomes meaningful within this causal setting. End-to-end delay includes signal acquisition, window formation, preprocessing, inference, decision logic, communication, and physical cue onset. A classifier may correctly describe the data from which its estimate was produced while that estimate becomes less relevant before the sound is delivered. Longer contextual windows can improve classification stability but also make the underlying evidence older. The appropriate latency threshold therefore cannot be assigned from computational speed alone; it must be evaluated relative to the temporal stability of the physiological condition being targeted.

SleepStim illustrates that peripheral sensing can meet at least some of these requirements in practice. Its smartwatch transmitted heart-rate and inertial data approximately once per second, and the system reported roughly 1 s latency between acquisition and availability of its N3 probability estimate. Cueing was suspended when signal transmission failed or the estimated probability of N3 fell below predefined thresholds (Whitmore et al. 2022). This is stronger evidence for closed-loop readiness than retrospective stage agreement alone because the physiological estimate actually governed intervention.

Wearable EEG has likewise been used within real-time home TMR. Salfi et al. used a wearable EEG headband to identify slow-wave activity and trigger vocabulary-associated cues during home sleep (Salfi et al. 2025). This demonstrates that electrophysiological information can also be incorporated into an autonomous home intervention. Neither study, however, establishes a universal architecture or a definitive performance threshold for future systems.

Uncertainty must also be treated as part of control rather than hidden behind a forced stage label. Signal degradation, state transitions, disagreement between signals, or low classifier confidence may leave the system without sufficient evidence to justify stimulation. An explicit abstention state is therefore preferable to mandatory classification when the cost of an inappropriate cue exceeds the cost of missing one opportunity.

Finally, the measurement task continues after stimulation. A cue may leave the physiological state largely unchanged, or it may produce movement, cortical activation, autonomic change, a transition to lighter sleep, or awakening. Post-cue information can therefore determine whether subsequent stimulation should continue, be delayed, or be suspended. Closed-loop TMR should be understood as a recurrent measurement–decision–intervention process rather than as a sleep-stage classifier connected directly to audio output.

5.6 Evidence Boundary for Sensing-Pathway Selection

The wearable literature supports three conclusions relevant to the translation problem. First, reduced-montage EEG can preserve substantial electrophysiological information and can support automated sleep staging outside full PSG, although signal quality and performance vary across devices, populations, and recording conditions (Arnal et al. 2020; Ravindran et al. 2025). Second, PPG and multimodal peripheral signals

Table 2 Operational distinction between retrospective sleep scoring, causal real-time state estimation, and closed-loop TMR.

Property	Retrospective scoring	Causal state estimation	Closed-loop TMR
Objective	Reconstruct sleep architecture after recording.	Estimate the current physiological state as sleep unfolds.	Use physiological evidence to authorize and adapt intervention.
Temporal context	Past and future epochs may be available.	Only previously acquired data are available.	Causal estimates are combined with cue history and subsequent physiological responses.
Latency	Generally not intervention-limiting.	Estimate must remain relevant to the current state.	Complete sensing-to-cue delay must preserve the validity of the intervention decision.
Uncertainty	Ambiguous epochs may be reconsidered retrospectively.	Confidence and signal quality must be available when the estimate is produced.	Insufficient evidence should result in delayed or withheld stimulation.
Signal loss	Corrupted intervals can be removed or reviewed later.	Reduces or prevents current-state estimation.	Should produce abstention until usable physiological evidence returns.
Post-cue monitoring	Not applicable.	Observation may continue without affecting an intervention.	Physiological responses influence whether subsequent cueing continues, changes, or stops.

Note. Closed-loop readiness is defined here as a translational operating requirement, not as a standardized clinical sleep-staging metric.

contain sufficient information to estimate broad sleep state and, in some systems, more detailed sleep architecture (Korkalainen et al. 2020; Chinoy et al. 2021; Svensson et al. 2024). Third, retrospective agreement with PSG is not equivalent to successful real-time intervention control.

The relevant comparison for TMR is therefore not “EEG accuracy versus PPG accuracy” in the abstract. Candidate pathways must instead be evaluated against the functions established in the preceding sections: recognition of physiologically supported cueing conditions, causal availability of the measurement, signal-quality awareness, bounded sensing-to-intervention delay, conservative handling of uncertainty, and detection of physiological disturbance following stimulation.

Existing home TMR studies demonstrate that both peripheral and EEG-based approaches can participate in automated cueing (Whitmore et al. 2022; Salfi et al. 2025). The available evidence does not, within this section alone, establish equivalence between them or determine which should serve as the first-generation implementation pathway. That decision requires comparison of evidentiary continuity, physiological observability, implementation uncertainty, and validation risk. Section 6 performs that translation using the requirements derived here.

6 Technology Translation: From Scientific Evidence to a First-Generation Reference Pathway

6.1 Translational Requirements Derived from the Evidence

The purpose of the present section is not to introduce additional biological claims, but to apply the evidence-to-engineering procedure defined in Section 2 to the findings established in Sections 3–5. The central translational question is whether the experimental conditions under which TMR has demonstrated measurable effects can be represented as system-level requirements without converting unresolved biological or technical assumptions into apparent capabilities. The seven general requirement categories defined in Section 2 are consolidated here into five intervention-specific requirements because measurement quality, intervention control, operational reliability, and empirical validation span multiple TMR-specific functions rather than forming independent implementation requirements.

Five requirements follow consistently from the preceding evidence.

First, the intervention requires **traceable memory preparation**. TMR depends on an association established before sleep, but the literature does not support a universal pre-sleep performance threshold below which cueing becomes invalid. Prior memory strength and the structure of cue–memory associations influence subsequent TMR effects in a condition-dependent manner (Creery et al. 2015; Cairney et al. 2016). A first-generation system should therefore preserve the identity of cue–memory mappings and record pre-sleep learning performance, while treating any numerical eligibility threshold as a parameter requiring prospective validation.

Second, the intervention requires **physiological eligibility**. Cue presentation should be restricted to sleep conditions supported by the TMR literature rather than governed solely by elapsed time. Meta-analytic evidence supports significant average effects during NREM stage 2 and slow-wave sleep, although neither state is universally optimal across all paradigms (Hu et al. 2020). An autonomous system must therefore estimate whether the current physiological interval is eligible for stimulation and must be capable of withholding intervention when that judgment is unsupported.

Third, the system requires **conservative disturbance control**. TMR depends on preserving the sleep physiology within which cueing occurs. Unsupervised and automated home studies indicate that stimulus intensity and sleep disturbance can materially influence behavioral outcome (Göldi and Rasch 2019; Whitmore et al. 2022). Cue delivery should therefore be governed by signal quality, physiological stability, recent stimulation history, and evidence of arousal or state transition. The literature does not establish universal thresholds for these variables; those thresholds remain implementation-specific validation questions.

Fourth, the intervention requires **temporal validity**. Physiological information used to authorize cueing must remain relevant when the cue is actually delivered. The system must therefore measure the delay between signal acquisition, state estimation, decision logic, scheduling, and physical cue onset. No single maximum acceptable latency can be derived from the present evidence because the relevant timescale depends on whether control is based on broad sleep stage, transient events, or oscillatory phase. End-to-end latency should consequently be treated as a measurable system

Table 3 Evidence-derived requirements for a first-generation closed-loop TMR system. The final column identifies assumptions that remain to be established prospectively rather than treated as settled design parameters.

Evidence basis	Translational implication	System-level requirement	Unresolved validation question
Cue-memory association	TMR operates on information associated with sensory cues before sleep.	Preserve cue-memory identity and record pre-sleep learning performance.	What encoding or retrieval criterion, if any, should determine eligibility for overnight cueing?
State-dependent efficacy	Cueing effects depend on the physiological context in which stimulation occurs.	Estimate intervention eligibility causally and permit abstention during unsupported or unstable states.	What state definition and confidence criterion provide adequate intervention reliability?
Sensitivity to disturbance	Excessive or poorly controlled stimulation may disrupt sleep and reduce or reverse behavioral benefit.	Control cue exposure and reassess physiological stability following stimulation.	Which post-cue signals and thresholds best identify biologically consequential disturbance?
Temporal dependence	The physiological evidence authorizing stimulation may become outdated before cue onset.	Measure and synchronize the complete sensing-to-stimulation pathway.	What end-to-end delay is acceptable for the selected level of physiological targeting?
Condition-dependent behavioral effect	System operation must be interpretable in relation to actual cueing conditions.	Maintain synchronized intervention records during experimental validation.	Which operational variables best predict successful versus unsuccessful TMR trials?

property whose acceptable range must be established relative to the physiological target.

Fifth, experimental validation requires **intervention traceability**. Delivered and withheld cues should be reconstructable in relation to the physiological evidence available at the time of the decision. At minimum, the validation system should retain synchronized information concerning cue identity, physiological state estimate, signal quality or confidence, decision time, actual cue onset, and subsequent evidence of sleep disruption. Behavioral outcomes can then be interpreted in relation to how the system actually operated rather than according to its intended protocol alone.

These requirements do not imply that every variable must ultimately appear in a deployed product. Some, particularly detailed intervention logging and cued-uncued behavioral comparisons, are requirements of scientific validation rather than permanent user-facing functionality. The purpose of the first-generation system is to make the intervention experimentally interpretable before optimization for usability or scale.

Table 3 summarizes the translation from scientific evidence to system requirement while retaining the principal uncertainty associated with each step.

6.2 Comparison of First-Generation Implementation Pathways

The requirements above constrain the implementation space but do not uniquely identify a device or architecture. Selection therefore requires comparison of candidate pathways according to two distinct questions: how strongly each pathway is connected

to the existing evidence, and how many additional assumptions must be validated before an unsuccessful or successful result can be interpreted.

Fixed-schedule versus physiology-guided cueing.

The relevant distinction is not between all “open-loop” and “closed-loop” TMR experiments. Many foundational studies achieved physiological control through researcher-supervised PSG rather than through an automated feedback algorithm. The translational problem arises when an autonomous system delivers cues according to elapsed time or a predetermined schedule without verifying that the sleeper remains in an appropriate state.

For an unsupervised system, physiology-guided control provides substantially greater interpretability. Current-state information can determine whether a cue is authorized, delayed, or withheld, while post-cue information can determine whether stimulation should continue. This introduces additional sensing and control complexity, but that complexity serves a scientific function: it preserves the relationship between the intervention and the physiological conditions under which TMR efficacy has been demonstrated. Closed-loop control is therefore selected here as a requirement of *autonomous state-dependent implementation*, not as a claim that TMR itself is invalid without an automated closed loop.

Wearable EEG versus peripheral physiological inference.

The sensing comparison requires particular caution because the available evidence supports both approaches, but at different levels of physiological observability.

Wearable EEG retains access to scalp electrophysiology used to characterize sleep state and selected events relevant to TMR. Reduced-montage systems have demonstrated substantial sleep-staging capability, although performance and signal usability vary across devices, populations, and recording conditions (Arnal et al. 2020; Ravindran et al. 2025). More importantly for the present translation problem, wearable EEG has already been incorporated into automated home TMR: Salfi et al. used real-time EEG-derived slow-wave detection to guide vocabulary cueing and reported a behavioral advantage for cued material (Salfi et al. 2025).

Peripheral sensing should not be characterized as experimentally unvalidated. PPG and related signals can provide substantial information about sleep state (Korkalainen et al. 2020), and SleepStim demonstrated that smartwatch-derived cardiac and inertial information can govern automated TMR cue delivery at home (Whitmore et al. 2022). Its behavioral results also demonstrated, however, the importance of stimulus intensity and intervention conditions. The current evidence therefore establishes peripheral closed-loop feasibility, but does not establish that peripheral-state estimates provide intervention control equivalent to electrophysiological monitoring across TMR protocols.

No direct head-to-head experiment currently demonstrates that an EEG-guided TMR controller produces greater behavioral efficacy than an otherwise equivalent peripheral controller. The choice between these pathways must therefore remain a translational inference rather than an experimentally established ranking. EEG offers closer continuity with the electrophysiological measurements used in much of the controlled TMR literature, whereas peripheral sensing reduces measurement burden at the cost of an additional inference between measured autonomic physiology and the cortical state relevant to the intervention.

Stage-aware versus event- or phase-aware targeting.

The literature also supports different levels of temporal specificity. Meta-analytic evidence demonstrates TMR benefits when cueing is restricted to broad NREM conditions, particularly N2 and slow-wave sleep (Hu et al. 2020). At finer temporal scales, spindle-related findings and closed-loop experiments indicate that the momentary oscillatory context of cue presentation can influence subsequent memory (Antony et al. 2018; Shimizu et al. 2018; Ngo and Staresina 2022; Nicolas et al. 2025).

Phase-specific control therefore has direct experimental support and should not be treated merely as a mechanistically plausible future extension. However, the available evidence remains narrower across memory domains, protocols, and independent replications than the evidence supporting TMR within broader NREM conditions. Moreover, existing phase-specific studies do not establish a universal incremental advantage over well-controlled stage-aware TMR. For example, Nicolas et al. found superior performance for UP- relative to DOWN-state cueing, but the UP-reactivated condition did not significantly outperform the non-reactivated condition (Nicolas et al. 2025).

Greater temporal specificity also introduces additional dependencies on real-time event detection, phase estimation, causal processing, synchronization, and physical cue timing. For the initial reference pathway, the relevant distinction is therefore not between a scientifically valid and an invalid strategy, but between two evidence-supported strategies with different levels of implementation dependency.

The first-generation pathway is consequently defined here as stage-aware and non-phase-locked. This choice should be interpreted as a sequencing decision, not as evidence that phase-aware control is ineffective or unnecessary. Fine-scale slow-wave or spindle information may be recorded during first-generation validation, while the incremental value of phase-aware control should subsequently be tested against the characterized reference configuration.

Existing-platform integration versus proprietary hardware.

The decision to begin with existing wearable EEG hardware follows a different type of reasoning. It is not a biological conclusion but an engineering sequencing judgment. Developing new sensing hardware simultaneously with a new closed-loop intervention would introduce additional uncertainty concerning electrode design, acquisition quality, synchronization, comfort, durability, and hardware reliability. Failure of the integrated experiment could then be difficult to localize.

Using existing hardware narrows this first validation problem only if the platform exposes the functions required by the intervention. A candidate device must provide sufficiently usable physiological signals in real time, reliable timestamps or synchronization, access compatible with causal processing, and a means of coordinating cue delivery with the recorded physiology. If no available platform satisfies these conditions, existing-platform integration cannot be assumed to be preferable simply because the hardware already exists.

The initial platform strategy should therefore be stated conditionally: development should preferentially integrate with a compatible and sufficiently validated wearable EEG platform rather than begin with proprietary hardware. The specific device

Table 4 Comparison of candidate implementation choices for a first-generation closed-loop TMR reference system. The table separates demonstrated evidence from uncertainties that would accompany each pathway.

Decision axis	Candidate options	Evidence-supported interpretation	First-generation implication
Cue control	Fixed-schedule autonomous cueing versus physiology-guided closed-loop control.	TMR efficacy is state-dependent; autonomous fixed scheduling cannot verify the physiology present at cue onset.	Use physiology-guided control with explicit ability to withhold stimulation.
Sensing	Peripheral physiological inference versus wearable EEG.	Both have participated in automated home TMR. Peripheral sensing infers neural state indirectly; EEG preserves scalp electrophysiological observability (Whitmore et al. 2022; Salfi et al. 2025).	Use wearable EEG as the initial reference pathway while retaining peripheral control as an experimentally viable alternative.
Temporal targeting	Stage-aware versus event- or phase-aware cueing.	NREM stage-dependent efficacy is established more broadly than any incremental benefit of precise phase targeting (Hu et al. 2020; Antony et al. 2018).	Begin with stage-aware, non-phase-locked control; test finer temporal targeting subsequently.
Platform	Compatible existing wearable EEG versus newly developed hardware.	No biological evidence requires proprietary hardware; new hardware would add an independent acquisition and validation problem.	Prefer existing-platform integration when real-time signal access, synchronization, and intervention control are technically available.

Note. The table does not rank technologies by universal quality. It identifies the pathway that introduces the fewest additional assumptions into the initial test of the closed-loop TMR intervention.

remains an engineering-selection question outside the evidentiary conclusion of this paper.

6.3 Evidence-Based Selection of the First-Generation Reference Pathway

The preceding comparison does not demonstrate that one architecture is universally superior. No study identified in the present synthesis directly compares complete EEG-guided and peripheral-guided TMR systems under identical participants, learning tasks, cueing policies, and outcome measures. The first-generation decision must therefore be understood as an *evidence-based translational judgment*: the preferred pathway is the one that allows the central intervention hypothesis to be tested while introducing the fewest additional unvalidated dependencies.

On this basis, the most defensible first-generation reference pathway is an **EEG-guided, stage-aware, real-time closed-loop TMR software layer intended for integration with compatible existing wearable EEG hardware**. This selection follows from the convergence of several observations rather than from any single study.

Controlled TMR evidence establishes that cueing efficacy depends on physiological state and produces a modest, condition-dependent behavioral effect (Hu et al. 2020). Wearable EEG has demonstrated that substantial sleep-related electrophysiological information can be retained in reduced, portable systems, while also revealing practical limitations related to signal quality and population generalization (Arnal et al. 2020; Ravindran et al. 2025). Most directly, EEG-guided automated TMR has produced a measurable vocabulary benefit under home conditions (Salfi et al. 2025). Together, these findings provide an empirical chain from laboratory physiology to wearable measurement to automated intervention.

Peripheral sensing remains scientifically credible. Sleep-stage inference from cardiac and multimodal peripheral signals is well supported, and SleepStim provides direct evidence that peripheral physiology can control automated home TMR (Korkalainen et al. 2020; Whitmore et al. 2022). The reason for not selecting it as the initial reference is therefore not absence of feasibility. Instead, a peripheral-first system would require the first validation program to establish both the TMR intervention and the adequacy of an indirect physiological control signal at the same time. EEG retains greater electrophysiological observability and allows failures in cueing, sleep-state detection, and post-stimulation response to be investigated more directly.

Stage-aware rather than phase-locked control follows the same sequencing principle. The behavioral evidence for NREM-targeted TMR is broader than the evidence that precise oscillatory-phase targeting yields additional behavioral benefit (Hu et al. 2020). Beginning with non-phase-locked control reduces the number of timing assumptions that must be validated simultaneously. Fine-scale slow-wave or spindle information remains scientifically valuable and may be recorded during first-generation experiments, but it should become a mandatory control variable only if prospective comparison demonstrates an advantage over the simpler reference policy.

Existing-platform integration similarly reduces the number of independent technical problems introduced into the first experiment. This choice remains conditional rather than absolute: an existing EEG device is suitable only if its signal access, timing behavior, data quality, and integration interfaces can satisfy the requirements established above. The paper therefore selects a *platform strategy*, not a specific commercial device.

The resulting decision can be expressed as a sequence of controlled uncertainty:

validated TMR evidence $\rightarrow$ electrophysiological state observation $\rightarrow$ causal eligibility decision $\rightarrow$ controlled cue delivery $\rightarrow$ post-cue observation $\rightarrow$ behavioral validation.

The value of the selected pathway is that each link can be observed and tested without simultaneously requiring a new sensing modality, a phase-sensitive controller, and a newly developed hardware platform to succeed.

This selection is deliberately revisable. Evidence that a peripheral system can reproduce cue-eligibility decisions, intervention timing, sleep stability, and behavioral outcomes relative to an EEG reference would weaken the rationale for retaining EEG in later generations. Similarly, a replicated behavioral advantage from event- or phase-aware control would justify increasing temporal specificity, while failure of existing wearable EEG platforms to provide adequate causal signal access or synchronization could justify dedicated hardware development.

The first-generation pathway should therefore not be interpreted as a claim that EEG is permanently superior to peripheral sensing, that stage-aware cueing is physiologically optimal, or that existing hardware will necessarily satisfy implementation requirements. It is a testable reference configuration chosen because the current evidence provides the clearest traceable path from established TMR physiology to an interpretable automated experiment. Detailed algorithm design, numerical operating thresholds, device selection, deployment format, and product validation remain subsequent research questions rather than conclusions of the present synthesis.

7 Limitations and Translational Boundaries

7.1 Limitations of the Evidence-Synthesis Process

The conclusions of this study are constrained first by the design of the evidence synthesis itself. The literature was identified through predefined translational research questions spanning memory neuroscience, sleep physiology, TMR, wearable sensing, real-time state estimation, and closed-loop intervention. This structure enabled evidence from otherwise separate fields to be evaluated within a common development problem, but it was not designed to provide exhaustive retrieval of every publication within each domain.

The resulting evidence base may therefore omit relevant work because of differences in terminology, indexing, disciplinary boundaries, publication access, or the progressive refinement of the research questions. This affects the comprehensiveness and reproducibility of the synthesis more directly than the validity of any single cited experiment. The pathway selected in Section 6 should consequently remain revisable if additional evidence materially changes the relative support for competing implementation strategies.

A second limitation is reviewer-dependent judgment. Source identification, critical appraisal, and translational interpretation were conducted by a single reviewer without duplicate screening, independent extraction, formal risk-of-bias scoring, or standardized certainty grading. The predefined research-question structure, explicit source-function criteria, preference for primary and independently validated evidence, and claim-level traceability described in Section 2 provided safeguards against unrestricted source selection, but they do not eliminate subjective judgment.

No new quantitative synthesis was performed. Effect estimates were taken from published meta-analyses or individual studies rather than recomputed across the assembled literature. The present study therefore cannot provide a new pooled estimate of TMR efficacy, formally rank the certainty of competing sensing pathways, or assign quantitative confidence to the pathway comparison. Its outputs should be interpreted as evidence-supported translational judgments rather than as systematic-review certainty estimates or independently pooled effect sizes.

7.2 Limitations of the Available Scientific Evidence

The TMR literature itself places important limits on translation. Experimental protocols vary in memory domain, initial learning strength, cue modality, stimulus intensity,

sleep stage, cue timing, physiological monitoring, participant characteristics, retention interval, and behavioral outcome. Accordingly, TMR should not be treated as a single standardized intervention. The meta-analysis by Hu et al. identified a positive average effect ($g = 0.29$) together with substantial between-study heterogeneity ($I^2 = 71\%$) (Hu et al. 2020). The same analysis identified funnel-plot asymmetry consistent with possible publication bias; under one trim-and-fill analysis, the estimated effect was reduced while remaining positive (Hu et al. 2020). The pooled value should therefore be interpreted as an estimate across heterogeneous experimental conditions rather than as an expected performance target for an engineered system.

External validity is also limited. Much of the evidence relevant to TMR derives from healthy participants completing tightly controlled memory tasks over one or a small number of nights. Such experiments are well suited to identifying causal effects, but they provide much less information about repeated unsupervised use, night-to-night reliability, long retention intervals, habituation to cues, adherence, age-related variation, atypical sleep architecture, or clinical populations. Existing home studies extend ecological validity but remain limited in number, scale, duration, and task diversity (Göldi and Rasch 2019; Whitmore et al. 2022; Salfi et al. 2025).

The long-term behavioral and physiological consequences of repeated TMR also remain insufficiently characterized. Short experimental protocols can determine whether appropriately delivered cues alter subsequent memory without establishing whether repeated intervention across weeks or months remains effective, produces habituation, changes normal sleep architecture, or affects memory processes beyond the intended material. The present evidence does not demonstrate such adverse effects, but neither does it justify treating their absence in short studies as evidence of long-term safety. Claims concerning durability, cumulative benefit, and repeated unsupervised exposure must therefore await longitudinal evaluation.

Mechanistic interpretation remains incomplete as well. Cue-evoked spindle activity, content-related EEG patterns, and temporal sensitivity to endogenous oscillatory dynamics are consistent with contemporary models of memory reactivation, but no single neurophysiological marker has been established as a validated surrogate for behavioral TMR success. Consequently, a system should not infer successful memory enhancement solely from detection of a spindle, slow wave, classifier output, or other physiological response.

The wearable literature introduces a further evidence boundary. Studies of wearable EEG, PPG, and consumer sleep-tracking systems differ substantially in population, hardware, signal preprocessing, class definitions, reference scoring, algorithm version, and validation setting (Arnal et al. 2020; Korkalainen et al. 2020; Chinoy et al. 2021; Ravindran et al. 2025). Performance reported for one device or algorithm cannot therefore be generalized to an entire sensing modality. Furthermore, much of the wearable evidence concerns retrospective agreement with PSG rather than causal operation within an intervention loop. Real-time TMR requires additional properties—including usable continuous signals, uncertainty handling, synchronization, bounded sensing-to-cue delay, abstention during unreliable intervals, and post-cue monitoring—that are rarely validated together in the same system.

7.3 Limitations of Translational Inference

The central translational limitation is that evidence for individual components does not constitute validation of their integration. The preceding sections provide support for several separate propositions: TMR can produce a modest memory-specific behavioral effect; sleep physiology can be observed through EEG and estimated through peripheral signals; reduced wearable systems can operate outside full PSG; and both EEG- and peripheral-guided systems have been used to automate cue delivery at home (Whitmore et al. 2022; Salfi et al. 2025). None of these findings individually demonstrates that the first-generation configuration selected in Section 6 will operate reliably as an integrated system.

Integration creates new failure modes at the interfaces among sensing, preprocessing, state estimation, control logic, communication, cue execution, and post-stimulation monitoring. A physiological classifier may perform adequately while the end-to-end intervention fails because of signal loss, timing error, synchronization failure, inappropriate thresholds, or an unrecognized post-cue transition. Conversely, a technically reliable closed-loop implementation may operate exactly as intended without producing a measurable behavioral benefit. Technical validity and biological efficacy must therefore be demonstrated separately before they can be interpreted jointly.

The EEG-first decision is also not the result of direct comparative validation. No evidence reviewed in this study establishes that an EEG-guided controller is behaviorally superior to an otherwise equivalent PPG- or multimodal-peripheral controller under matched participants, learning tasks, cue policies, and physiological conditions. Similarly, the evidence does not establish that stage-aware stimulation is superior to phase-aware control, or that integration with existing hardware is superior to proprietary hardware in all relevant dimensions.

The selected reference pathway is consequently a sequencing decision based on evidentiary continuity and the number of additional assumptions introduced into the initial validation problem. EEG was selected because it preserves closer access to the electrophysiological domain used to characterize sleep and has already supported automated home TMR, not because permanent superiority over peripheral sensing has been demonstrated. Stage-aware control was selected because its behavioral evidence base is broader and its implementation requires fewer timing assumptions than phase-locked control, not because it has been shown to be physiologically optimal.

The translation from scientific evidence to system requirements is itself qualitative. Requirements such as physiological eligibility, conservative abstention, temporal validity, and post-cue monitoring follow from the reviewed evidence, but the literature does not provide universal numerical thresholds for implementing them. The present analysis therefore identifies *what must be validated* more strongly than it determines *the numerical values at which validation will succeed*.

7.4 Boundaries of the Present Conclusions

Within these limitations, the evidence supports controlled development and prospective evaluation of an EEG-guided, stage-aware, real-time closed-loop TMR reference

pathway. It does not establish that the resulting system will produce reliable memory benefits across participants, tasks, recording nights, devices, or broader populations. It also does not establish that an integrated system can yet maintain sufficient signal quality, state-estimation reliability, timing accuracy, disturbance control, or behavioral efficacy during extended unsupervised operation.

The present study does not provide validated numerical thresholds for sleep-state confidence, signal quality, cue intensity, cue spacing, state stability, acceptable latency, or post-stimulation arousal. These variables should remain configurable during experimental development and should be determined through platform-specific physiological and behavioral validation rather than fixed from indirect literature estimates.

Nor does the paper establish a finalized system architecture, production hardware configuration, deployable software product, clinical intervention, or consumer-ready memory-enhancement technology. The selected configuration is a *first-generation reference pathway*: a testable arrangement chosen to provide an interpretable starting point for engineering and experimental validation.

The conclusions can therefore be stated at four different levels. The **biological evidence** supports TMR as a condition-dependent means of influencing sleep-dependent memory processing. The **experimental evidence** supports a modest average behavioral effect under appropriate conditions. The **translational analysis** supports EEG-guided, stage-aware closed-loop control as the most defensible initial reference pathway under the evidence reviewed. The **product-level evidence**, however, remains insufficient because the proposed integrated system has not yet been implemented and prospectively validated.

These boundaries are not peripheral qualifications to the central conclusion; they define its valid scope. The present work narrows the development space, identifies the physiological and measurement functions that should be preserved, and establishes a reference configuration against which subsequent engineering choices can be tested. The remaining uncertainties therefore become explicit research questions rather than hidden assumptions, providing the basis for the validation priorities described in Section 8.

8 Future Directions

8.1 Prospective Validation of the First-Generation Reference Pathway

The immediate research priority is not expansion of the proposed system, but prospective validation of the first-generation reference pathway identified in Section 6. The central question is whether an EEG-guided, stage-aware, closed-loop TMR system can reproduce the physiological control and measurable behavioral effect required for the translational argument to remain valid. Validation should therefore address two related but analytically distinct problems: whether the intervention is executed as intended and whether correctly executed intervention produces the expected memory effect.

Intervention fidelity.

The first requirement is to establish that closed-loop operation can be reconstructed objectively from synchronized physiological and intervention records. Validation

should determine whether usable EEG can be maintained during overnight recording, whether state estimates are generated causally, whether low-quality or ambiguous intervals lead to abstention, and whether actual cue onset occurs while the physiological evidence supporting the decision remains valid. Post-cue monitoring should determine whether stimulation is followed by arousal, stage transition, signal degradation, or other changes relevant to subsequent cueing.

These outcomes should be evaluated at the intervention level rather than only through whole-night summary statistics. For each potential cueing opportunity, the system should permit reconstruction of the physiological evidence available at the time, the resulting control decision, whether a cue was actually delivered, and the physiological state surrounding cue onset. Such intervention-level traceability would distinguish failure of sensing or control from absence of a biological TMR effect.

No universal thresholds for state confidence, signal quality, latency, cue spacing, or arousal detection are established by the present synthesis. Initial studies should therefore estimate these operating ranges empirically rather than embedding literature-derived values as fixed specifications. The objective is not to maximize stimulation frequency, but to identify operating conditions under which intervention decisions remain physiologically interpretable.

Behavioral efficacy under verified closed-loop operation.

Behavioral validation should use controlled paradigms in which pre-sleep learning, cue-memory associations, cue exposure, physiological state at stimulation, and post-sleep retention can be linked at the item or trial level. Cued and uncued material should be appropriately matched so that any observed difference can be distinguished from ordinary sleep-dependent consolidation or general practice effects.

The principal question is not merely whether mean memory performance improves, but whether a reproducible TMR effect is observed when the system demonstrably executes the intended protocol. Given the modest and heterogeneous effect reported in the broader literature (Hu et al. 2020), a single positive experiment would provide insufficient evidence for reference-system validation. Replication across participants, recording nights, and preferably independent experimental samples should establish whether the behavioral effect persists when intervention fidelity is controlled.

Technical and behavioral outcomes need not be evaluated in completely separate studies, but behavioral efficacy should not be interpreted independently of technical fidelity. A null behavioral result obtained during unreliable state estimation or mistimed cue delivery would not constitute a clean test of TMR. Conversely, repeated null effects obtained despite verified physiological operation would challenge the assumption that the selected reference pathway preserves the intervention conditions necessary for measurable benefit.

This distinction creates an explicit falsification criterion for the first-generation hypothesis. If reliable EEG acquisition, causal state-dependent control, and disturbance-limited cue delivery can be demonstrated but replicated behavioral benefit cannot, development should not progress automatically toward greater algorithmic complexity. The scientific premise, task selection, cueing protocol, or practical value of the pathway would instead require re-evaluation.

8.2 Identifying the Physiological Variables That Improve Control

Once the reference pathway can be evaluated reproducibly, the next question is which additional physiological information actually improves intervention outcomes. Future development should distinguish variables that are merely measurable from variables that provide causal or predictive value for deciding when and how to stimulate.

Adaptive control as a testable hypothesis.

Individual and night-to-night differences in sleep architecture, sensory threshold, pre-sleep memory strength, cue history, and response to previous stimulation provide plausible inputs for adaptive control. However, adding adaptive variables increases model flexibility and creates opportunities for overfitting, unstable decision policies, and post-hoc explanations of successful or unsuccessful trials.

Each adaptive feature should therefore be introduced through prospective comparison with a simpler reference policy. A variable should influence cue eligibility, intensity, spacing, repetition, or temporary suspension only when its inclusion improves at least one predefined outcome: behavioral efficacy, preservation of sleep, intervention precision, or reliability of abstinence. Prediction of outcome within the same dataset is not sufficient evidence that a variable should control future stimulation.

Where personalization is investigated, validation should distinguish population-level predictive effects from individual-specific adaptation. Apparent subject-specific patterns may reflect noise when only a small number of nights or cueing events are available. Adaptive policies should therefore be evaluated prospectively and, where feasible, against fixed-policy or cross-validated baselines before being interpreted as genuine personalization.

Stage-aware versus event- and phase-aware targeting.

A separate question concerns the temporal resolution at which TMR should operate. The rationale for finer targeting is supported not only by the association of memory processing with slow oscillations and sleep spindles, but also by closed-loop experiments demonstrating phase-dependent behavioral and electrophysiological effects (Staresina et al. 2015; Cairney et al. 2018; Antony et al. 2018; Shimizu et al. 2018; Ngo and Staresina 2022; Nicolas et al. 2025). These findings establish phase-aware TMR as an experimentally supported alternative rather than a purely theoretical extension. They do not yet establish that phase-aware stimulation provides a robust and generalizable incremental advantage over well-controlled stage-aware cueing across memory domains and implementation conditions.

This question should therefore be tested directly rather than resolved through mechanistic plausibility. Stage-aware and temporally refined policies should be compared under matched learning tasks, sensing conditions, cue exposure, and outcome measures. Evaluation should include behavioral effect size, trigger precision, missed opportunities, end-to-end timing error, artifact susceptibility, sleep disruption, and reproducibility across participants and nights.

Fine temporal control should be adopted only if its additional physiological specificity translates into a reproducible advantage that exceeds the cost of greater detection and synchronization complexity. A failure to demonstrate such an advantage would

itself be informative: it would support retaining stage-aware control as the more parsimonious intervention strategy rather than treating increased temporal precision as inherently superior.

8.3 Comparative Sensing and Sensor-Reduction Experiments

The EEG-first decision made in Section 6 should ultimately be tested rather than preserved by assumption. Once a sufficiently characterized EEG-guided reference configuration exists, it can provide the experimental basis for determining how much physiological sensing is actually necessary for effective TMR.

Hybrid sensing and complementary physiological information.

Initial comparative studies should acquire EEG and peripheral signals synchronously rather than replacing EEG immediately. Candidate peripheral signals include PPG, accelerometry, respiratory information, and other measures that may contribute information about movement, arousal, sleep stability, or signal reliability.

The relevant question is not whether peripheral sensors improve agreement with conventional sleep-stage labels. Their value should be measured by whether they improve intervention decisions. Examples include identifying intervals in which EEG is unreliable, detecting physiological disturbance following stimulation, supporting abstention decisions, or reducing uncertainty about state transitions.

A hybrid configuration would therefore be justified when the additional signal provides measurable information beyond the EEG reference that improves intervention reliability, behavioral outcome, or preservation of sleep. Conversely, a sensor should not be retained merely because its data can be acquired.

Reduced EEG and peripheral-dominant control.

Sensor reduction should be evaluated prospectively as an intervention problem rather than solely as a sleep-staging problem. Reduced-channel EEG, EEG-peripheral hybrid systems, and eventually PPG-dominant controllers should be compared with the reference configuration using synchronized recordings and, where possible, identical cueing opportunities.

Relevant endpoints include agreement in cue-eligibility decisions, false authorization of stimulation, abstention during uncertain intervals, sensing-to-cue timing, detection of post-cue disturbance, preservation of sleep continuity, and behavioral TMR outcome. A reduced-sensor system that reproduces retrospective stage labels but systematically changes which cues are delivered has not demonstrated equivalence at the intervention level.

Peripheral control should therefore not be required to reproduce every EEG feature. The more meaningful question is whether the reduction in neural observability produces an unacceptable loss of intervention validity. A peripheral or hybrid pathway could justify progression if it preserves behavioral efficacy and physiological safety within prospectively defined bounds while materially reducing measurement burden.

This creates a testable route toward later-generation sensing rather than a permanent EEG dependency. If peripheral signals can reproduce the decisions and outcomes required for effective closed-loop TMR, the rationale for retaining EEG would weaken. Conversely, systematic disagreement in physiologically important intervals would identify functions for which direct electrophysiological observation remains necessary.

8.4 Generalization, Longitudinal Evaluation, and Responsible Translation

Demonstration of a reference-system effect would establish neither broad cognitive enhancement nor readiness for deployment. The final research phase must determine whether the intervention survives changes in task, population, time scale, and operating environment without losing efficacy, interpretability, or acceptable effects on sleep.

Generalization across memory domains.

TMR effects should be evaluated separately across memory systems and learning tasks rather than generalized from a single successful paradigm. Verbal paired associates, spatial learning, procedural skills, emotional memories, and more complex educational material differ in encoding structure, neural dependence, cue representation, and behavioral outcome. The effect observed in one domain should therefore be treated as evidence for that experimental class until cross-domain replication demonstrates broader applicability.

Generalization studies should preserve task-appropriate controls and report both absolute learning performance and the differential effect of cueing. Transfer to uncued but related knowledge should be measured explicitly rather than assumed. Evidence that TMR strengthens one set of cued items does not by itself support claims of general improvement in learning ability, intelligence, or memory capacity.

Population heterogeneity and individual response.

Future studies should progressively include participants differing in age, baseline memory performance, sleep architecture, sensory threshold, and other characteristics likely to influence intervention response. Expansion to clinical populations should occur only when the intended therapeutic question is independently justified and when the physiological and behavioral outcome measures are appropriate to that population.

Group-level efficacy should not be treated as evidence of predictable individual benefit. Future trials should characterize the distribution of responses, including reproducible non-response and possible adverse-response patterns, and should distinguish stable individual differences from night-to-night variability.

Longitudinal efficacy and sleep integrity.

Repeated-night studies are necessary to determine whether TMR effects persist, accumulate, diminish, or vary across extended use. Relevant outcomes include retention over longer intervals, night-to-night effect stability, cue habituation, changes in sensory responsiveness, sleep continuity, stage distribution, arousal burden, and effects on uncued material.

Potential outcomes such as habituation, progressive sleep fragmentation, or interference with other memory processes should be treated as hypotheses to be measured rather than consequences assumed to occur. Longitudinal validation should therefore evaluate both efficacy and the preservation of normal sleep physiology under repeated exposure.

A sustained intervention should be considered scientifically supported only when behavioral benefit remains reproducible without evidence that the procedure progressively compromises the physiological state on which the intervention depends.

Responsible translation and evidentiary claims.

The ethical problem surrounding closed-loop TMR extends beyond data protection because the system intervenes in memory processing during a state in which the user cannot actively supervise each decision. Future studies should therefore address not only informed consent and protection of physiological data, but also control over which memories are eligible for cueing, transparency regarding automated intervention, the ability to suspend or withdraw participation, and the distinction between experimentally demonstrated effects and speculative enhancement claims.

EEG recordings, sleep profiles, learning performance, and intervention histories should be collected according to data-minimization principles and protected according to the sensitivity and intended use of the information. Expansion toward educational, occupational, clinical, or enhancement contexts should not be treated as equivalent because the acceptable evidence, risk, and oversight requirements differ across those applications.

Communication of efficacy should remain proportional to the evidence. A modest average TMR effect should not be represented as guaranteed improvement, and successful performance in a constrained task should not be advertised as general memory enhancement. Particular caution is required before research is extended to populations with reduced capacity to consent or to applications in which participation could become coercive.

The long-term research objective is therefore not simply to produce a smaller, more adaptive, or more scalable TMR system. It is to determine the minimal physiological information and intervention complexity required to produce a reproducible memory effect while preserving sleep and maintaining an interpretable relationship between measurement, stimulation, and outcome. Progression beyond the first-generation reference pathway should occur only when an alternative demonstrates a measurable improvement in efficacy, reliability, safety, or measurement burden without introducing greater unresolved uncertainty.

9 Conclusion

This study asked how current evidence on sleep-dependent memory consolidation, Targeted Memory Reactivation (TMR), and wearable sleep monitoring can be translated into a scientifically defensible first-generation pathway for real-time closed-loop TMR. The evidence supports a qualified but actionable answer. TMR is no longer only a biologically plausible hypothesis: controlled experiments and meta-analytic evidence demonstrate a modest, condition-dependent effect on subsequent memory, while home-based studies show that physiologically guided cueing can be automated outside the conventional sleep laboratory (Hu et al. 2020; Whitmore et al. 2022; Salfi et al. 2025). At the same time, the literature does not establish a mature memory-enhancement technology, a universally effective cueing protocol, or an integrated system whose reliability has been demonstrated across users, tasks, devices, and repeated nights.

The principal translational conclusion is therefore the selection of an **EEG-guided, stage-aware, real-time closed-loop TMR software layer, intended for integration with compatible existing wearable EEG hardware, as the first-generation reference pathway**. This selection is not based on evidence that

EEG is permanently superior to peripheral sensing. Peripheral physiology has already supported automated home TMR, and later reduced-sensor or hybrid approaches remain scientifically credible. EEG is selected first because it currently preserves closer continuity with the electrophysiological domain used to characterize sleep and TMR, while avoiding the additional inference required by peripheral-only sensing. Stage-aware rather than phase-locked control follows the same reasoning: the evidence for state-dependent NREM cueing is broader than the evidence that finer oscillatory targeting produces a reproducible incremental behavioral advantage. Integration with existing hardware similarly represents a sequencing decision intended to avoid combining intervention validation with unnecessary hardware-development uncertainty.

The broader contribution of this work is not the specification of a finished device, algorithm, or product architecture, but the construction of a traceable evidence-to-engineering argument. Biological findings were translated into functional implications; those implications were converted into system-level requirements; candidate implementation pathways were then compared according to physiological observability, temporal suitability, evidence maturity, and the additional assumptions required for validation. This process identified a critical distinction that extends beyond TMR: the existence of individually feasible components does not demonstrate the validity of their integration. Retrospective sleep-stage accuracy does not establish intervention readiness, technical execution does not establish behavioral efficacy, and engineering convenience does not substitute for physiological justification. A defensible neurotechnology pathway must preserve these distinctions rather than resolve scientific uncertainty through design choice alone.

The resulting pathway therefore remains a testable reference configuration, not a validated technology. Prospective work must establish that usable EEG can be maintained during overnight operation, that physiological eligibility can be estimated causally with appropriate handling of uncertainty, that cue delivery remains temporally linked to the state that authorized it, that stimulation does not systematically destabilize sleep, and that technically faithful operation produces a reproducible behavioral effect. Failure at that final step would be scientifically meaningful: if the system reliably reproduces the intended physiological intervention yet repeated controlled studies fail to reproduce a memory benefit, increasing algorithmic complexity would not by itself justify continued translation. The intervention hypothesis, task conditions, or practical significance of the pathway would require reconsideration.

Accordingly, the endpoint of this study is neither a claim of deployment readiness nor an argument for technological finality. It is a narrowing of the translational problem to a configuration in which the remaining uncertainties can be observed, tested, and potentially falsified. If validated, the EEG-guided reference pathway could provide the experimental baseline against which adaptive control, finer temporal targeting, multimodal sensing, reduced-channel EEG, and peripheral-only approaches are evaluated. The central outcome is therefore a scientifically bounded route from evidence to experiment: one that defines not only what should be investigated first, but also what evidence must exist before the next technological claim is justified.

Declarations

Funding

The author received no funding, grants, or other financial support for the conduct of this research or the preparation of this manuscript.

Competing Interests

The author declares no current financial interests, patents, patent applications, commercial agreements, or affiliations with organizations that could benefit financially from the publication of this work. The author intends to explore the potential future development and commercialization of technologies related to the research direction described in this manuscript. At the time of submission, no company, commercial entity, or proprietary technology associated with this work has been established.

Ethics Approval

Not applicable. This study is a literature-based translational evidence synthesis and did not involve the recruitment of human participants, the collection of new human participant data, or research involving animals.

Consent to Participate

Not applicable. No human participants were recruited for this study.

Consent for Publication

Not applicable. The manuscript contains no identifiable personal information or participant-level material requiring consent for publication.

Data Availability

No new empirical datasets were generated or analyzed in this study. The scientific evidence synthesized in this work was derived from the published literature cited in the manuscript.

Materials Availability

Not applicable. No unique experimental materials were generated or used in this study.

Code Availability

Not applicable. No original software, computational model, or analysis code was developed or used to generate the findings reported in this study.

Author Contributions

V.G. conceived the study, formulated the research questions, conducted the literature identification and critical review, developed the evidence-to- engineering methodology,

performed the evidence synthesis and translational analysis, developed the conceptual figures and tables, and wrote and revised the manuscript. The author approved the final manuscript and takes responsibility for its scientific content.

Acknowledgements

The author used AI Assistant during manuscript preparation for assistance with manuscript organization and language refinement. All scientific claims, interpretations, references, and final text were independently reviewed and verified by the author, who takes full responsibility for the content of the manuscript.

References

- Antony JW, Piloto L, Wang M, et al (2018) Sleep spindle refractoriness segregates periods of memory reactivation. *Current Biology* 28(11):1736–1743.e4. <https://doi.org/10.1016/j.cub.2018.04.020>
- Arnal PJ, Thorey V, Debellemanniere E, et al (2020) The dream headband compared to polysomnography for electroencephalographic signal acquisition and sleep staging. *Sleep* 43(11):zsaa097. <https://doi.org/10.1093/sleep/zsaa097>
- Buzsáki G, Draguhn A (2004) Neuronal oscillations in cortical networks. *Science* 304(5679):1926–1929. <https://doi.org/10.1126/science.1099745>
- Buzsáki G, Anastassiou CA, Koch C (2012) The origin of extracellular fields and currents—EEG, ECoG, LFP and spikes. *Nature Reviews Neuroscience* 13(6):407–420. <https://doi.org/10.1038/nrn3241>
- Cairney SA, Lindsay S, Sobczak JM, et al (2016) The benefits of targeted memory reactivation for consolidation in sleep are contingent on memory accuracy and direct cue-memory associations. *Sleep* 39(5):1139–1150. <https://doi.org/10.5665/sleep.5772>
- Cairney SA, Guttesen A, El Marj N, et al (2018) Memory consolidation is linked to spindle-mediated information processing during sleep. *Current Biology* 28(6):948–954.e4. <https://doi.org/10.1016/j.cub.2018.01.087>
- Chinoy ED, Cuellar JA, Huwa KE, et al (2021) Performance of seven consumer sleep-tracking devices compared with polysomnography. *Sleep* 44(5):zsaa291. <https://doi.org/10.1093/sleep/zsaa291>
- Creery JD, Oudiette D, Antony JW, et al (2015) Targeted memory reactivation during sleep depends on prior learning. *Sleep* 38(5):755–763. <https://doi.org/10.5665/sleep.4670>
- Diekelmann S, Born J (2010) The memory function of sleep. *Nature Reviews Neuroscience* 11(2):114–126. <https://doi.org/10.1038/nrn2762>

- Dudai Y, Karni A, Born J (2015) The consolidation and transformation of memory. *Neuron* 88(1):20–32. <https://doi.org/10.1016/j.neuron.2015.09.004>
- Fernandez LMJ, Lüthi A (2020) Sleep spindles: Mechanisms and functions. *Physiological Reviews* 100(2):805–868. <https://doi.org/10.1152/physrev.00042.2018>
- Göldi M, Rasch B (2019) Effects of targeted memory reactivation during sleep at home depend on sleep disturbances and habituation. *npj Science of Learning* 4:5. <https://doi.org/10.1038/s41539-019-0044-2>
- Grant MJ, Booth A (2009) A typology of reviews: An analysis of 14 review types and associated methodologies. *Health Information & Libraries Journal* 26(2):91–108. <https://doi.org/10.1111/j.1471-1842.2009.00848.x>
- Hu X, Cheng LY, Chiu MH, et al (2020) Promoting memory consolidation during sleep: A meta-analysis of Targeted Memory Reactivation. *Psychological Bulletin* 146(3):218–244. <https://doi.org/10.1037/bul0000223>
- Klinzing JG, Niethard N, Born J (2019) Mechanisms of systems memory consolidation during sleep. *Nature Neuroscience* 22(10):1598–1610. <https://doi.org/10.1038/s41593-019-0467-3>
- Korkalainen H, Aakko J, Duce B, et al (2020) Deep learning enables sleep staging from photoplethysmogram for patients with suspected sleep apnea. *Sleep* 43(11):zsaa098. <https://doi.org/10.1093/sleep/zsaa098>
- Ngo HVV, Staresina BP (2022) Shaping overnight consolidation via slow-oscillation closed-loop targeted memory reactivation. *Proceedings of the National Academy of Sciences of the United States of America* 119(44):e2123428119. <https://doi.org/10.1073/pnas.2123428119>
- Nicolas J, King BR, Lévesque D, et al (2025) Unraveling the neurophysiological correlates of phase-specific enhancement of motor memory consolidation via slow-wave closed-loop targeted memory reactivation. *Nature Communications* 16:2644. <https://doi.org/10.1038/s41467-025-57602-2>
- Oudiette D, Paller KA (2013) Upgrading the sleeping brain with Targeted Memory Reactivation. *Trends in Cognitive Sciences* 17(3):142–149. <https://doi.org/10.1016/j.tics.2013.01.006>
- Rasch B, Born J (2013) About sleep’s role in memory. *Physiological Reviews* 93(2):681–766. <https://doi.org/10.1152/physrev.00032.2012>
- Rasch B, Büchel C, Gais S, et al (2007) Odor cues during slow-wave sleep prompt declarative memory consolidation. *Science* 315(5817):1426–1429. <https://doi.org/10.1126/science.1138581>

- Ravindran KKG, Della Monica C, Atzori G, et al (2025) Evaluation of dreem headband for sleep staging and EEG spectral analysis in people living with alzheimer’s disease and older adults. *Sleep* 48(8):zsaf122. <https://doi.org/10.1093/sleep/zsaf122>
- Rosenberg RS, Van Hout S (2013) The american academy of sleep medicine inter-scorer reliability program: Sleep stage scoring. *Journal of Clinical Sleep Medicine* 9(1):81–87. <https://doi.org/10.5664/jcsm.2350>
- Rudoy JD, Voss JL, Westerberg CE, et al (2009) Strengthening individual memories by reactivating them during sleep. *Science* 326(5956):1079. <https://doi.org/10.1126/science.1179013>
- Salfi F, D’Atri A, Arnone B, et al (2025) Promoting vocabulary learning during sleep at home using closed-loop Targeted Memory Reactivation. *Journal of Sleep Research* 34(6):e70000. <https://doi.org/10.1111/jsr.70000>
- Scammell TE, Arrigoni E, Lipton JO (2017) Neural circuitry of wakefulness and sleep. *Neuron* 93(4):747–765. <https://doi.org/10.1016/j.neuron.2017.01.014>
- Shimizu RE, Connolly PM, Cellini N, et al (2018) Closed-loop targeted memory reactivation during sleep improves spatial navigation. *Frontiers in Human Neuroscience* 12:28. <https://doi.org/10.3389/fnhum.2018.00028>
- Silber MH, Ancoli-Israel S, Bonnet MH, et al (2007) The visual scoring of sleep in adults. *Journal of Clinical Sleep Medicine* 3(2):121–131. <https://doi.org/10.5664/jcsm.26814>
- Snyder H (2019) Literature review as a research methodology: An overview and guidelines. *Journal of Business Research* 104:333–339. <https://doi.org/10.1016/j.jbusres.2019.07.039>
- Squire LR, Genzel L, Wixted JT, et al (2015) Memory consolidation. *Cold Spring Harbor Perspectives in Biology* 7(8):a021766. <https://doi.org/10.1101/cshperspect.a021766>
- Staresina BP, Bergmann TO, Bonnefond M, et al (2015) Hierarchical nesting of slow oscillations, spindles and ripples in the human hippocampus during sleep. *Nature Neuroscience* 18(11):1679–1686. <https://doi.org/10.1038/nn.4119>
- Svensson T, Madhawa K, Nt H, et al (2024) Validity and reliability of the Oura Ring Generation 3 (Gen3) with Oura Sleep Staging Algorithm 2.0 (OSSA 2.0) when compared to multi-night ambulatory polysomnography: A validation study of 96 participants and 421,045 epochs. *Sleep Medicine* 115:251–263. <https://doi.org/10.1016/j.sleep.2024.01.020>
- Wang B, Antony JW, Lurie S, et al (2019) Targeted memory reactivation during sleep elicits neural signals related to learning content. *The Journal of Neuroscience*

39(34):6728–6736. <https://doi.org/10.1523/JNEUROSCI.2798-18.2019>

Whitmore NW, Harris JC, Kovach T, et al (2022) Improving memory via automated Targeted Memory Reactivation during sleep. *Journal of Sleep Research* 31(6):e13731. <https://doi.org/10.1111/jsr.13731>

de Zambotti M, Cellini N, Goldstone A, et al (2019) Wearable sleep technology in clinical and research settings. *Medicine & Science in Sports & Exercise* 51(7):1538–1557. <https://doi.org/10.1249/MSS.0000000000001947>